# Bioprocess Engineering

Career Avenues

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# KINETICS OF MICROBIAL GROWTH, SUBSTRATE UTILIZATION AND PRODUCT FORMATION

## MICROBIAL GROWTH

The growth of a microorganism may result in the production of a range of metabolites but to produce a particular metabolite the desired organism must be grown under precise cultural conditions at a particular growth rate. If a microorganism is introduced into a nutrient medium that supports its growth, the inoculated culture will pass through a number of stages and the system is termed batch culture. Initially, growth does not occur and this period is referred to as the lag phase and may be considered a period of adaptation. Following an interval during which the growth rate of the cells gradually increases, the cells grow at a constant, maximum rate and this period is referred to as the log or exponential phase, which may be described by the equation.

$$\frac{dx}{dt} = \mu x \quad \text{---------- (1)}$$

Where, x is the cell concentration (mg $ml^{-1}$), t is the time of incubation (h), and μ the specific growth rate ($h^{-1}$). On integration equation (1) gives

$$x_t = x_o e^{\mu t} \quad \text{---------- (2)}$$

Where $x_o$ is the cell concentration at time zero and $x_t$ is the cell concentration after a time interval, t (h).

Thus, a plot of the natural logarithm of the cell concentration against time gives a straight line, the slope of which equals the specific growth rate. The specific growth rate during the exponential phase is the maximum for the prevailing conditions and is described as the maximum specific growth rate $\mu_{max}$. Equations (1) and (2) ignore the facts that growth results in the depletion of nutrients and the accumulation of toxic by-products and thus predict that growth continues indefinitely.

However, in reality, as substrate (nutrient) is exhausted and toxic products accumulate, the growth rate of the cells deviates from the maximum and eventually growth ceases and the culture enters the stationary phase. After a further period of time, the culture enters the death phase and the number of viable cells declines. This classic representation of microbial growth is illustrated in Figure 1.

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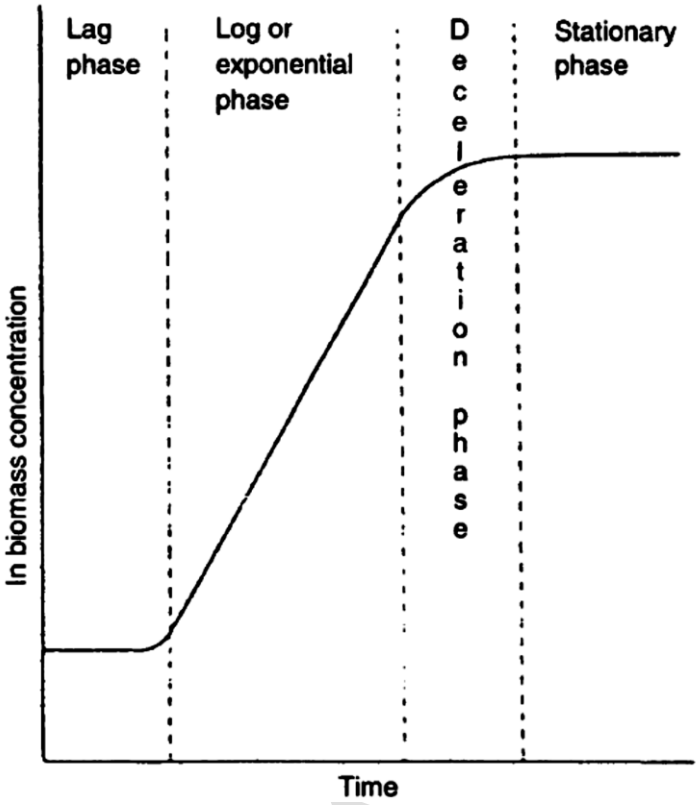



Figure 1: Growth of a 'typical' microorganism under batch culture conditions

It should be remembered that this description refers to the behaviour of growth may be extended by addition of an aliquot of fresh medium to the vessel. If the fresh medium is added continuously, at an appropriate rate, and the culture vessel is fitted with an overflow device, such that culture is displaced by the incoming fresh medium, a continuous culture may be established. The growth of the cells in a continuous culture of this type is controlled by the availability of the growth limiting chemical component of the medium and thus, the system is described as a chemostat. In this system a steady-state is eventually achieved and the loss of biomass via the overflow is replaced by cell growth. The flow of medium through the system is described by the term dilution rate, D, which is equal to the rate of addition of medium divided by the working volume of the culture vessel. The balance between growth of cells and their loss from the system may be described as

$$\frac{dx}{dt} = growth - output$$

Or

$$\frac{dx}{dt} = \mu x - Dx$$

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Under steady-state conditions,

$$\frac{dx}{dt} = 0$$

therefore, $\mu x = Dx$ and $\mu = D$.

Hence, the growth rate of the organisms is controlled by the dilution rate, which is an experimental variable. It will be recalled that under batch culture conditions an organism will grow at its maximum specific growth rate and, therefore, it is obvious that a continuous culture may be operated only at dilution rates below the maximum specific growth rate. Thus, within certain limits, the dilution rate may be used to control the growth rate of a chemostat culture.

The mechanism underlying the controlling effect of the dilution rate is essentially the relationship between μ,specific growth rate, and s, the limiting substrate concentration in the chemostat, demonstrated by Monod' in 1942:

$\mu = \mu_{max} S/ (K_s+S)$ ----------(3)

where $K_s$, is the utilization or saturation constant, which is numerically equal to the substrate concentration when μ is half $\mu_{max}$. At steadystate, $\mu = D$, and, therefore,

$$D = \mu_{max}\bar{s}/(K_s + \bar{s})$$

Where S(s bar) is the steady-state concentration of substrate in the chemostat, and

$$\bar{s} = K_s D/(\mu_{max} - D)$$ ---------- (4)

Equation (4) predicts that the substrate concentration is determined by the dilution rate. In effect, this occurs by growth of the cells depleting the substrate to a concentration that supports that growth rate equal to the dilution rate. If substrate is depleted below the level that supports the growth rate dictated by the dilution rate the following sequence of events takes place:

- The growth rate of the cells will be less than the dilution rate and they will be washed out of the vessel at a rate greater than they are being produced, resulting in a decrease in biomass concentration.
- The substrate concentration in the vessel will rise because fewer cells are left in the vessel to consume it.
- The increased substrate concentration in the vessel will result in the cells growing at a rate greater than the dilution rate and biomass concentration will increase.
- The steady state will be re-established.

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- Thus, a chemostat is a nutrient-limited self-balancing culture system that may be maintained in a steady state over a wide range of sub-maximum specific growth rates.

Fed-batch culture is a system that may be considered intermediate between batch and continuous processes. The term fed-batch is used to describe batch cultures that are fed continuously or sequentially, with fresh medium without the removal of culture fluid. Thus, the volume of a fed-batch culture increases with time. Pirt described the kinetics of such a system as follows. If the growth of an organism were limited by the concentration of one substrate in the medium the biomass at stationary phase, $x_{max}$, would be described by the equation:

$$x_{max} = YS_R$$

where Y is the yield factor and is equal to the mass of cells produced per gram of substrate consumed and $S_R$ is the initial concentration of the growth limiting substrate. If fresh medium were to be added to the vessel at a dilution rate less than $\mu_{max}$ then virtually all the substrate would be consumed as it entered the system:

$$FS_R = \mu(X/Y)$$

where, F is the flow rate and X is the total biomass in the vessel, i.e. the cell concentration multiplied by the culture volume.

Although the total biomass (X) in the vessel increases with time the concentration of cells, x, remains virtually constant; thus $dx/dt = 0$ and $\mu = D$. Such a system is then described as quasi-steady-state. As time progresses and the volume of culture increases, the dilution rate decreases. Thus, the value of D is given by the expression

$$D = F/(V_0 + F_t)$$

where F is the flow rate, *Vo* is the initial culture volume, and t is the time.

Monod' kinetics predicts that as ***D*** falls residual substrate concentration should also decrease, resulting in an increase in biomass. However, over the range of growth rates operating the increase in biomass should be insignificant. The major difference between the steady state of the chemostat and the quasi-steady-state of a fed-batch culture is that in a chemostat D (hence, μ) is constant whereas in a fed-batch system D (hence, μ) decreases with time. The dilution rate in a fed-batch system could be kept constant by increasing exponentially, the flow rate using a computer-control system.

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# STERILIZATION OF AIR AND MEDIA

## MEDIUM STERILISATION

As pointed out by Corbett (1985), media may be sterilized by filtration, radiation, ultrasonic treatment, chemical treatment or heat. However, for practical reasons, steam is used almost universally for the sterilization of fermentation media. The major exception is the use of filtration for the sterilization of media for animal cell culture such media are completely soluble and contain heat labile components making filtration the method of choice. Filtration techniques will be considered later in this chapter. Before the techniques which are used for the steam sterilization of culture media are discussed it is necessary to discuss the kinetics of sterilization. The destruction of micro-organisms by steam (moist heat) may be described as a first-order chemical reaction and thus, may be represented by the following equation:

$$-\mathrm{d}N/\mathrm{d}t = kN \qquad \text{----------(5.1)}$$

where N - is the number of viable organisms present, t-is the time of the sterilization treatment, k-is the reaction rate constant of the reaction, or the specific death rate.

It is important at this stage to appreciate that we are considering the total number of organisms present in the volume of medium to be sterilized, not the concentration - the minimum number of organisms to contaminate a batch is one, regardless of the volume of the batch. On integration of equation (5.1) the following expression is obtained:

$$N_t/N_0 = \mathrm{e}^{-kt} \qquad \text{----------(5.2)}$$

where No is the number of viable organisms present at the start of the sterilization treatment, Nt is the number of viable organisms present after a treatment period, t. On taking natural logarithms, equation (5.2) is reduced to:

$$\ln\left(N_t/N_0\right) = -kt \qquad \text{----------(5.3)}$$

The graphical representations of equations (5.1) and (5.3) are illustrated in Fig. 5.1, from which it may be seen that viable organism number declines exponentially over the treatment period. A plot of the natural logarithm of $N_t/N_o$ against time yields a straight line, the slope of which equals - k.

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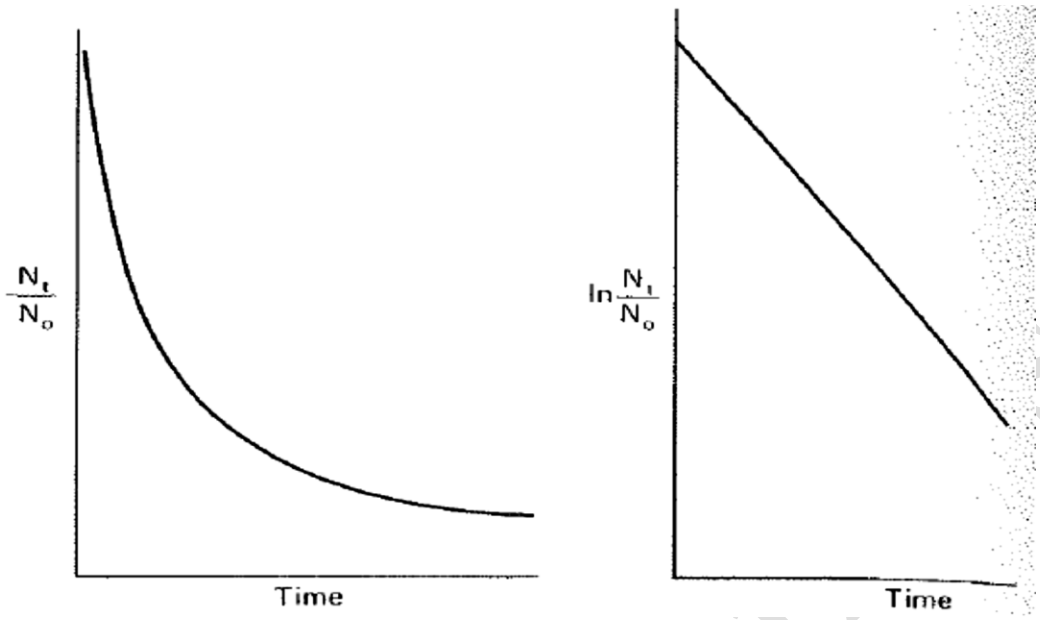



Figure. 5.1. Plots of the proportion of survivors and the natural logarithm of the proportion of survivors in a population of microorganisms subjected to a lethal temperature over a time period.

This kinetic description makes two predictions which appear anomalous:

An infinite time is required to achieve sterile conditions (i.e. $N_t = 0$).

After a certain time there will be less than one viable cell present.

Thus, in this context, a value of $N_t$ of less than one is considered in terms of the probability of an organism surviving the treatment. For example, if it were predicted that a particular treatment period reduced the population to 0.1 of a viable organism, this implies that the probability of one organism surviving the treatment is one in ten. This may be better expressed in practical terms as a risk of one batch in ten becoming contaminated.

As with any first-order reaction, the reaction rate increases with increase in temperature due to an increase in the reaction rate constant, which, in the case of the destruction of microorganisms, is the specific death rate (k). Thus, k is a true constant only under constant temperature conditions. The relationship between temperature and the reaction rate constant was demonstrated by Arrhenius and may be represented by the equation:

$$\mathrm{d} \ln k / \mathrm{d}T = E / RT^2 \quad \text{----------(5.4)}$$

where, E is the activation energy, R is the gas constant, *T* is the absolute temperature.

On integration equation (5.4) gives:

$$k = Ae^{-E/RT} \quad \text{----------(5.5)}$$

where, *A* is the Arrhenius constant.

On taking natural logarithms, equation (5.5) becomes:

$$\ln k = \ln A - E/RT. \quad \text{----------(5.6)}$$

From equation (5.6) it may be seen that a plot of ln *k* against the reciprocal of the absolute temperature will give a straight line. Such a plot is termed an Arrhenius plot and enables the calculation of the activation energy and the prediction of the reaction rate for any temperature. By combining together equations (5.3) and (5.5), the following expression may be derived for the heat sterilization of a pure culture at a constant temperature:

$$\ln N_0/N_t = A \cdot t \cdot e^{-E/RT}. \quad \text{----------(5.7)}$$

Deindoerfer and Humphrey (1959) used the term $\ln N_o/N_t$ as a design criterion for sterilization, which has been variously called the Del factor, Nabla factor and sterilization criterion represented by the term "∇". Thus, the Del factor is a measure of the fractional reduction in viable organism count produced by a certain heat and time regime. Therefore:

$$\nabla = \ln (N_0/N_t)$$

but $$\ln(N_0/N_t) = kt$$

and $$kt = A \cdot t \cdot e^{-(E/RT)}$$

thus $$\nabla = A \cdot t \cdot e^{-(E/RT)}. \qquad (5.8)$$

On rearranging, equation (5.8) becomes:

$$\ln t = E/RT + \ln (\nabla/A). \quad \text{----------(5.9)}$$

Thus, a plot of the natural logarithm of the time required to achieve a certain '∇' value against the reciprocal of the absolute temperature will yield a straight line, the slope of which is dependent on the activation energy, as shown in Fig. 5.2.

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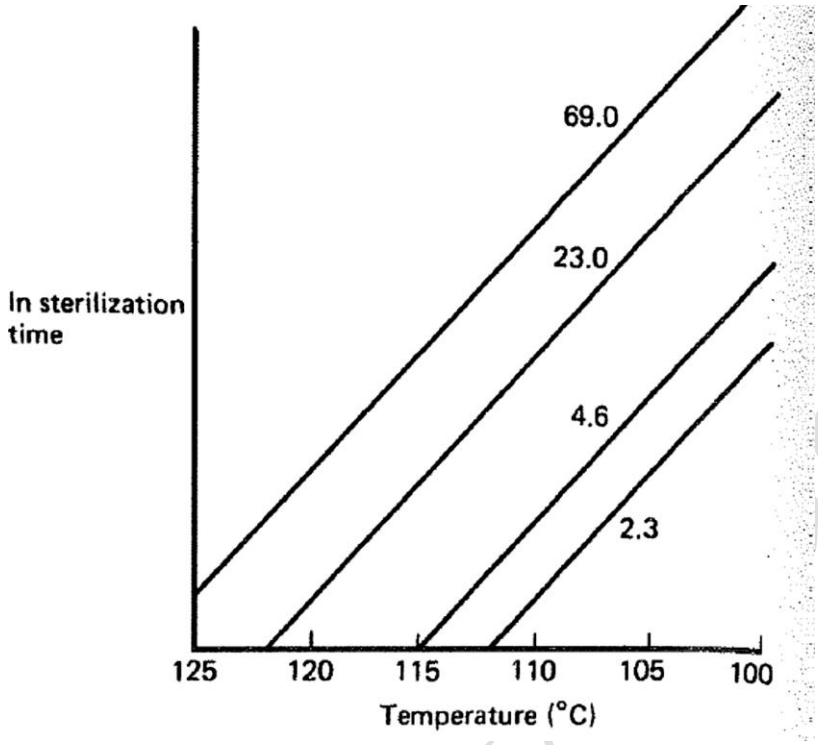



Fig. 5.2. The effect of sterilization and temperature on the Del factor achieved in the process.

The figures on the graph indicate the Del factors for each straight line.

From Fig. 5.2 it is clear that the same degree of sterilization (∇) may be obtained over a wide range of time and temperature regimes; that is, the same degree of sterilization may result from treatment at a high temperature for a short time as from a low temperature for a long time.

This kinetic description of bacterial death enables the design of procedures for the sterilization of fermentation broths. By choosing a value for $N_t$, procedures may be designed having a certain probability of achieving sterility, based upon the degree of risk that is considered acceptable.

According to Deindoerfer and Humphrey (1959), Richards (1968), Banks (1979) and Corbett (1985) a risk factor of one batch in a thousand being contaminated is frequently used in the fermentation industry that is, the final microbial count in the medium after sterilization should be $10^{-3}$ viable cells. However, to apply these kinetics it is necessary to know the thermal death characteristics of all the taxa contaminating the fermenter and unsterile medium. This is an impossibility and therefore, the assumption may be made that the only microbial contaminants present are spores of *Bacillus stearothelmophilus*, one of the most heat-resistant microbial types known. Thus, by adopting *B. stearothennophilus* as the design organism a considerable safety factor should be built into the calculations. It should be remembered that *B. Stearothennophilus* is not always adopted as the design organism. If the most heat-resistant organism contaminating the medium ingredients is known, then it may be advantageous to base the sterilization process on this organism. Deindoerfer and Humphrey (1959) determined the thermal death characteristics of *B. stearothermophilus* spores are:

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Activation energy =67.7 kcal $mole^{-1}$

Arrhenius constant = 1 X 1036.2 $second^{-1}$

∇ In overall can be calculated using equation:

$$\nabla_{overall} = \nabla_{heating} + \nabla_{holding} + \nabla_{cooling}.$$

Two types of reaction contribute to the loss of nutrient quality during sterilization:

**(I) Interactions between nutrient components of the medium**

A common occurrence during sterilization is the Maillard-type browning reaction which results in discoloration of the medium as well as loss of nutrient quality. These reactions are normally caused by the reaction of carbonyl groups, usually from reducing sugars, with the amino groups of amino acids and proteins. Problems of this type are normally resolved by sterilizing the sugar separately from the rest of the medium and recombining the two after cooling.

**(II) Degradation of heat labile components**

Certain vitamins, amino acids and proteins may be degraded during a steam sterilization regime. In extreme cases, such as the preparation of media for animal-cell culture, filtration may be used and this aspect will be discussed later in the chapter. However, for the vast majority of fermentations these problems may be resolved by the judicious choice of steam sterilization regime.

**Advantages of continuous sterilization over batch sterilization:**

- Superior maintenance of medium quality.
- Ease of scale-up.
- Easier automatic control.
- The reduction of surge capacity for steam.
- The reduction of sterilization cycle time.
- Under certain circumstances, the reduction of fermenter corrosion

**Advantages of batch sterilization over continuous sterilization:**

- Lower capital equipment costs.
- Lower risk of contamination –continuous processes require the aseptic transfer of the sterile broth to the sterile vessel.
- Easier manual control.
- Easier to use with media containing a high proportion of solid matter.

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## METHODS OF BATCH STERILIZATION

The batch sterilization of the medium for fermentation may be achieved either in the fermentation vessel or in a separate mash cooker. Richards (1966) considered the relative merits of in situ medium sterilization and the use of a special vessel. The major advantages of a separate medium sterilization vessel may be summarized as:

One cooker may be used to serve several fermenters and the medium may be sterilized as the fermenters are being cleaned and prepared for the next fermentation, thus saving time between fermentations.

The medium may be sterilized in a cooker in a more concentrated form than would be used in the fermentation and then diluted in the fermenter with sterile water prior to inoculation. This would allow the construction of smaller cookers.

In some fermentations, the medium is most viscous during sterilization and the power requirement for agitation is not alleviated by aeration, as it would be during the fermentation proper. Thus, if the requirement for agitation during in situ sterilization were removed, the fermenter could be equipped with a less powerful motor. Obviously, the sterilization kettle would have to be equipped with a powerful motor, but this would provide sterile medium for several fermenters. The fermenter would be spared from the corrosion that may occur with medium at high temperature.

**The major disadvantages of a separate medium sterilization vessel may be summarized as:**

- The cost of constructing a batch medium sterilizer is much the same as that for the fermenter.
- If a cooker serves a large number of fermenters, complex pipe work would be necessary to transport the sterile medium with the inherent dangers of contamination.
- Mechanical failure in a cooker supplying medium to several fermenters would render all the fermenters temporarily redundant. The provision of contingency equipment may be prohibitively costly.

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## THE DESIGN OF CONTINUOUS STERILIZATION PROCESS

The design of continuous sterilization cycles may be approached in exactly the same way as for batch sterilization systems. The continuous system includes a time period during which the medium is heated to the sterilization temperature, a holding time at the desired temperature and a cooling period to restore the medium to the fermentation temperature. The temperature of the medium is elevated in a continuous heat exchanger and is then maintained in an insulated serpentine holding coil for the holding period. The length of the holding period is dictated by the length of the coil and the flow rate of the medium. The hot medium is then cooled to the fermentation temperature using two sequential heat exchangers, the first utilizing the incoming medium as the cooling source (thus conserving heat by heating-up the incoming medium) and the second using cooling water. The major advantage in the continuous process is that a much higher temperature may be utilized, thus reducing the holding time and reducing the degree of nutrient degradation. The required Del factor may be achieved by the combination of temperature and holding time, which gives an acceptably small degree of nutrient decay. Richards (1968) quoted the following example to illustrate the range of temperature-time regimes which may be employed to achieve the same probability of obtaining sterility. The Del factor for the example sterilization was 45.7 and the following temperature time regimes were calculated to give the same Del factor:

| Temperature | Holding time |
|---|---|
| 130° | 2.44 minutes |
| 135° | 51.9 seconds |
| 140° | 18.9 seconds |
| 150° | 2.7 seconds |

Furthermore, because a continuous process involves treating small increments of medium the heating-up and cooling-down periods are very small compared with those in a batch system. There are two types of continuous sterilizer which may be used for the treatment of fermentation media: the indirect heat exchanger and the direct heat exchanger (steam injector). The most suitable indirect heat exchangers are of the double-spiral type which consists of two sheets of high-grade stainless steel which have been curved around a central axis to form a double spiral. The ends of the spiral are sealed by covers.

To achieve sterilization temperatures steam is passed through one spiral and medium through the other in countercurrent streams. Spiral heat exchangers are also used to cool the medium after passing through the holding coil. Incoming unsterile medium is used as the cooling agent in the first cooler so that the incoming medium is partially heated before it reaches the sterilizer and, thus, heat is conserved.

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**The major advantages of the spiral heat exchanger are:**

The two streams of medium and cooling liquid, or medium and steam, are separated by a continuous stainless steel barrier with gasket seals being confined to the joints with the end plates. This makes cross contamination between the two streams unlikely.

The spiral route traversed by the medium allows sufficient clearances to be incorporated for the system to cope with suspended solids. The exchanger tends to be self-cleaning which reduces the risk of sedimentation, fouling and burning-on.

## Indirect Plate Heat Exchangers

Indirect plate heat exchangers consist of alternating plates through which the countercurrent streams are circulated. The plates are separated by gaskets and failure of these gaskets can cause cross contamination between the two streams. In addition, the clearances between the plates are such that suspended solids in the medium may block the exchanger and thus, the system is useful only in sterilizing completely soluble media. However, the plate exchanger is more adaptable than the spiral system in that extra plates may be added to increase its capacity. The continuous steam injector injects steam directly into the unsterile broth.

The Advantages And Disadvantages Of The System

- Very short (almost instantaneous) heating up times.
- It may be used for media containing suspended solids.
- Low capital cost.
- Easy cleaning and maintenance.
- High steam utilization efficiency.
-

However, the disadvantages are:

- Foaming may occur during heating.
- Direct contact of the medium with steam requires that allowance be made for condense dilution and requires clean steam, free from anticorrosion additives.

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## FILTER STERILIZATION OF AIR

Aerobic fermentations require the continuous addition of considerable quantities of sterile air. Although it is possible to sterilize air by heat treatment, the most commonly used sterilization process is filtration. Fixed pore filters (which have an absolute rating) are very widely used in the fermentation industry and several manufacturers produce filtration systems for air sterilization. These systems, like those for the sterilization of liquids, consist of pleated membrane cartridges designed to be accommodated in stainless steel modules.

The most common construction material used for the pleated membranes for air sterilization is PTFE, which is hydrophobic and is therefore resistant to wetting. Also, PTFE filters may be steam sterilized and are resistant to ammonia which may be injected into the air stream, prior to the filter, for pH control. As was seen for the filter sterilization of liquids it is essential that a prefilter is incorporated up-stream of the absolute filter. The prefilter traps large particles such as dust, oil and carbon from the compressor and pipescale and rust from the pipe work. The use of a coalescing prefilter also ensures the removal of water from the air; entrained water is coalesced in the filter (air flow being from the inside of the filter to the outside) and is discharged via an automatic drain.

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# BATCH, FED-BATCH AND CONTINUOUS PROCESSES

## BATCH REACTOR

If the bioreactor is only fed once and no more feeding after that, we say that, the bioreactor is operating on batch mode. The bioreactor will be allowed to run till completion. At the end of the rinthe fermentation is terminated before a new fermentation is initiated again.

In the real situation this is difficult to achieve as a true batch reactor since in a batch reactor there must not be any input or withdrawal from the bioreactor until the fermentation is completed. In most aerobic batch reactor air or oxygen is continuously sparged in or the addition of some other minor nutrients. If we do not supply oxygen into the aerobic bioreactor then the bioreactor will turn anaerobic and that will be the end of the aerobic batch bioreactor. There should not also be any removals of samples from the bioreactor throughout the fermentation run as in the case of off line monitoring and sampling.

## CONTINUOUS BIOREACTOR

On the other hand, if the bioreactor is fed continuously then the bioreactor is termed a continuous bioreactor. Under such situation if the continuous bioreactor is a fixed volume bioreactor, the amount of feed introduced into the bioreactor is equivalent to the spent volume removed from the bioreactor.

Despite the ideal characteristics of the continuous bioreactor, the process itself is sensitive and subjected to influence from various factors such as risks of contaminations, cell or biomass washouts and changes in the biotic phase of the bioreactor.

## FED BATCH BIOREACTOR

Fed Batch is the intermediary model of bioreactor operation. This fed batch mode is characterized by predetermined or controlled addition of nutrients into the bioreactor at certain times of fermenter operations. This mode of feeding nutrients into the fed batch bioreactor will allow temporal variations in the supply of nutrients. The beautiful thing about fed batch operation of a bioreactor is it allows us a degree of control on the process and operations of the fed batch bioreactor. We can control the rate of growth of the microorganisms or the concentration of the biomass by controlling parameters such as frequency of feeds, hydraulic loadings and concentrations of feed.

It is used in industry to maintain high concentration of microorganisms and reduce feedback inhibition especially catabolite repression. Fed batch is the most popular mode in fermentation industries.

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## THE MODE OF OPERATION OF FERMENTATION PROCESSES

Microorganisms may be grown in batch, fed-batch, or continuous culture, and continuous culture offers the most control over the growth of the cells. However, the commercial adoption of continuous culture is confined to the production of biomass to a limited extent, in the production of potable and industrial alcohol. The superiority of continuous culture for biomass production is overwhelming, as may be seen from the following account, but for other microbial products the disadvantages of the system outweigh the improved process control which the technique offers.

Productivity in batch culture may be described by the equation.

$$R_{batch} = (x_{max} - x_0)/(t_i + t_{ii}) \quad \text{---------- (6)}$$

where,

- $R_{batch}$ is the output of the culture in terms of biomass concentration per hour
- $x_{max}$, is the maximum cell concentration achieved at stationary phase
- $x_o$ is the initial cell concentration at inoculation, $t_i$, is the time during which the organism grows at $\mu_{max}$ and
- $t_{ii}$ is the time during which the organism is not growing at $\mu_{max}$ and includes the lag phase, the deceleration phase, and the periods of batching, sterilizing and harvesting.

The productivity of a continuous culture may be represented as

$$R_{cont} = D\bar{x}(1 - t_{iii}/T) \quad \text{---------- (7)}$$

where,

- $R_{cont}$ is the output of the culture in terms of cell concentration per hour,
- $t_{iii}$ is the time period prior to the establishment of a steady-state and includes time for vessel preparation, sterilization and operation in batch culture prior to continuous operation,
- ***T*** is the time period during which steady-state conditions prevail, and
- X(x bar) is the steady state cell concentration.

Maximum output of biomass per unit time (i.e**.** productivity) in a chemostat may be achieved by operating at the dilution rate giving the highest value of DX(x bar), this value being referred to as $D_{max}$.Batch fermentation productivity, as described by equation (6), is an average for the total time of the fermentation. Because $dx/dt=\mu x$, the productivity of the culture increases with time and, thus, the vast majority of the biomass in a batch process is produced near the end of the log phase. In a steady-state chemostat, operating at, or near, $D_{max}$, the productivity remains constant,

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and maximum, for the whole fermentation. In addition, a continuous process may be operated for a very long time so that the nonproductive period, $\mathbf{t_{iii}}$ in equation (7), may be insignificant. However, the non-productive time element for a batch culture is a very significant period, especially as the fermentation would have to be re-established many times during the running time of a comparable continuous process and, therefore, $\mathbf{t_{ii}}$**;** would be recurrent. The steady-state nature of a continuous process is also advantageous in that the system should be far easier to control than a comparable batch one. During batch fermentation, heat output, acid or alkali production, and oxygen consumption will range from very low rates at the start of the fermentation to very high rates during the late logarithmic phase. Thus, the control of the environment of such a system is far more difficult than that of a continuous process where, at steady state, production and consumption rates are constant. Furthermore, a continuous process should result in a more constant labour demand than a comparable batch one.

A frequently quoted disadvantage of continuous systems is their susceptibility to contamination by foreign organisms. The prevention of contamination is essentially a problem of fermenter design, construction, and operation and should be overcome by good engineering and microbiological practice. ICI recognized the overwhelming advantages of a continuous biomass process and overcame the problems of contamination by building a secure fermenter capable of very long periods of aseptic operation. The production of growth-associated by-products, such as ethanol, should also be more efficient in continuous culture. However, continuous brewing has met with only limited success and UK breweries have abandoned such systems owing to problems of flavour and lack of flexibility. The production of industrial alcohol, on the other hand, should not be limited by the problems encountered by the brewing industry and continuous culture should be the method of choice for such a process.

The adoption of continuous culture for the production of biosynthetic (as opposed to catabolic) microbial products has been extremely limited. Although, theoretically, it is possible to optimize a continuous system such that optimum productivity of a metabolite should be achieved, the long-term stability of such systems is precarious, owing to the problem of strain degeneration.

A consideration of the kinetics of continuous culture reveals that the system is highly selective and will favour the propagation of the best-adapted organism in a culture. Best-adapted in this context refers to the affinity of the organism for the limiting substrate at the operating dilution rate.

A commercial organism is usually highly mutated such that it will produce very high amounts of the desired product. Therefore, in physiological terms, such commercial organisms are extremely inefficient and a revertant strain, producing less of the desired product, may be better adapted to the cultural conditions than the superior producer and will come to dominate the culture. This phenomenon, termed by Calott as contamination from within, is the major reason for the lack of use of continuous culture for the production of microbial metabolites.

Although the fermentation industry has been reluctant to adopt continuous culture for the production of microbial metabolites, very considerable progress has been made in the development of fed-batch systems. Fed-batch culture may be used to achieve a considerable

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degree of process control and to extend the productive period of a traditional batch process without the inherent disadvantages of continuous culture described previously. The major advantage of feeding a medium component to a culture, rather than incorporating it entirely in the initial batch, is that the nutrient may be maintained at a very low concentration during the fermentation.

A low (but constantly replenished) nutrient level may be advantageous in:

- Maintaining conditions in the culture within the aeration capacity
- Removing the repressive effects of medium components such as
- Avoiding the toxic effects of a medium component.

The earliest example of the commercial use of fed-batch culture is the production of bakers' yeast. It was recognized as early as 1915 that an excess of malt in the production medium would result in a high rate of biomass production and an oxygen demand which could not be met by the fermenter. This resulted in the development of anaerobic conditions and the formation of ethanol at the expense of biomass. The solution to this problem was to grow the yeast initially in a weak medium and then add additional medium at a rate less than the organism could use it. It is now appreciated that a high glucose concentration represses respiratory activity, and in modern yeast production plants the feed of molasses is under strict control based on the automatic measurement of traces of ethanol in the exhaust gas of the fermenter. As soon as ethanol is detected the feed rate is reduced. Although such systems may result in low growth rates, the biomass yield is near that theoretically obtainable.

The penicillin fermentation provides a very good example of the use of fed-batch culture for the production of a secondary metabolite. The penicillin process is a two-stage fermentation; an initial growth phase is followed by the production phase or idiophase. During the production phase glucose is fed to the fermentation at a rate which allows a relatively high growth rate (and therefore rapid accumulation of biomass) yet maintains the oxygen demand of the culture within the aeration capacity of the equipment. If the oxygen demand of the biomass were to exceed the aeration capacity of the fermenter anaerobic conditions would result and the carbon source would be used inefficiently. During the production phase the biomass must be maintained at a relatively low growth rate and, thus, the glucose is fed at a low dilution rate. Phenylacetic acid is a precursor of the penicillin molecule but it is also toxic to the producer organism above a threshold concentration. Thus, the precursor is also fed into the fermentation continuously, thereby maintaining its concentration below the inhibitory level.

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# AERATION AND AGITATION

## AERATION

### AIR INLET SPARGERS

Gases for the maintenance of dissolved oxygen, and in the case of mammalian culture, pH, need to be introduced into the vessel, and this is done in two ways – either by sparging into the liquid phase or by overlaying gas into the headspace. When sparging into the vessel this happens just below the impellors. All the gas is filtered prior to entry into the bioreactor by passing it through a 0.22-μm filter. Gas requirements are illustrated in the table below.

Table

| Gas | Cell culture | Microbial culture | Cell culture | Microbial culture |
|---|---|---|---|---|
| | Sparging | | Overlay | |
| Air | Approx. 0.1 vvm | 1–2 vvm | 0.1 vvm | 10% of 'air to sparger' |
| $O_2$ | 10% of 'air' | 20–30% of 'air' | NA | NA |
| $CO_2$ | 10–25% of 'air' | 20–30% of 'air' | 10% of 'air to sparger' | NA |
| $N_2$ | 10–25% of 'air' | 20–30% of 'air' | NA | NA |

vvm = volume per (working) volume per minute.

Gas that is sparged in to the vessel will enter the vessel though several different types of sparge line, again dependent on the type of organism being grown. To minimize the risk of contamination a filter is always placed prior to the sparge line. The first illustrated (Fig.1a) is the conventional 'L' design, for high gas flow rates, typically up to 2 vessel volumes per minute. Gas is dispersed through precision drill holes in the base of the pipe, and the resulting gas bubbles driven in to the liquid phase by the agitation and baffling systems.

The alternative illustrated (Fig. 1b) is the 'microporous' design, usually for low gas flow rates, typically up to 0.1 vessel volumes per minute. Gas is pushed through a fine sintered tip. Each pore is approx. 15 μm and so gas enters the liquid phase in a very fine dispersion of high surface area, thus improving mass transfer. This sparge design is sometimes favoured for cell culture as the bubbles are smaller and so contain less energy.

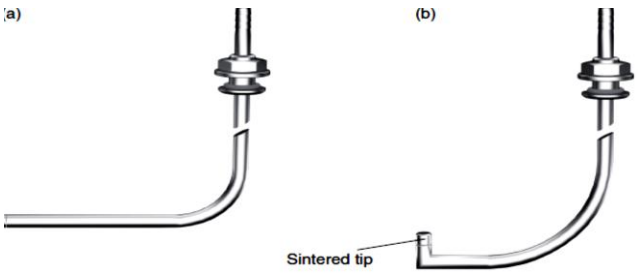



Figure. 1a -b**:** Example of spargers. (a) L sparger; (b) sintered sparger

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## EXHAUST GAS CONDENSER

The gas that has not been utilised by the process will pass through and exit the vessel. This gas will collect moisture as it goes through the medium so to prevent excessive evaporation and the blocking of the off gas filter a condenser is placed prior to the filter. The condenser is of baffled design, thus increasing the surface area and condensate return to the vessel. Chilled water is circulated round the condenser to maximize the condensing of gas moisture so that it is returned to the vessel. The cold water source should be a consideration as is it no longer environmentally or economically acceptable to leave a tap running to drain. A chiller unit should be used in this way, the environmental impact in minimized and the temperature of the water is constant. The same chiller could be used to supply a heat exchanger to remove the excess heat generated during an exothermic fermentation.

# AGITATION

### BAFFLES

Baffles are flat paddle like structures, typically 10% of the vessel internal diameter. As so often in fermentation, the question arises as to 'Why the magic number? – 10 in this case'. Winkler (1990) showed that wider baffles have little effect, while with narrower baffles the baffling effect decreases rapidly. Baffles sit inside the vessel next to the wall and are used to break the rotational flow of liquid in the vessel that may occur at high agitation rates. The use of baffles eliminates laminar flow, creates a turbulent flow and enhances mass transfer. Baffles tend only to be used in bacterial/yeast systems. It should be remembered that other vessel fittings, such as pH / $DO_2$ probes and heat exchangers that go into the vessel, will have a baffling effect. Mammalian systems tend not to use baffles, as the shear forces created behind a baffle may be sufficient to damage the cells.

## DRIVES/COUPLING

### IMPELLOR TYPES

All bioreactors require mixing to some extent. The mixing achieved by the impellors will be required to perform a number of different functions. These will include maintaining a uniform environment in the vessel in terms of heat, nutrient, and if present, solid particle distribution, fluid and gas mixing, and air/oxygen dispersion. All these things must be achieved efficiently, but without damage to the cells being grown.

To achieve these goals, several different types of impellor may be used. The choice of impellor is dictated by the physical robustness of the cells being grown, and the need for oxygen transfer. A typical mammalian culture will use a marine impellor (Fig. 2) that will be half the diameter of the vessel.

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Figure 2: Marine Impellor

This impellor will be either scoping, which means it creates an upward current in the vessel, or overtaxing which means it creates a downward flow in the vessel. These impellors are used specifically due to the low sheer forces created when they are rotating thus minimizing damage to the cells while, at the same time, maintaining a homogeneous mix with in the vessel. Rushton impellors (Fig.3) are typically used in bacterial and yeast fermentations.

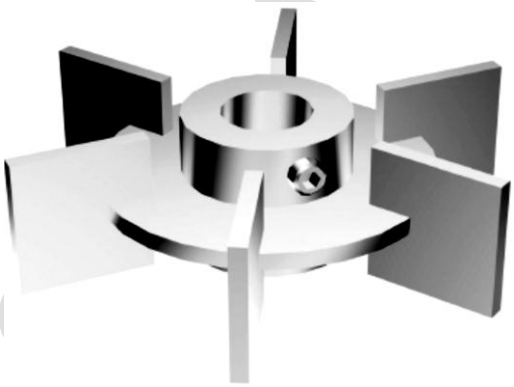


Figure 3: Rushton impellor

These impellors will be one-third of the diameter of the vessel, although for viscous cultures (filamentous fungi or Streptomycete cultures) a 0.5 diameter Rushton turbine might be used. The stirring speeds that Rushton impellors are used at can be up to ten-times those of the marines. This is possible as microbial cells are far more robust than mammalian cells. This speed of mixing is essential in a bacterial fermentation, because as well as mixing, the Rushton impellors are also helping to maintain the dissolved oxygen, and so meet the oxygen demands of the bacterial fermentation in a far more effective way than do the marine impellors. They do this by breaking up the gas bubbles as they hit the impellors thus increasing the surface area of the bubbles leading to greater gas transfer.

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The maintenance of dissolved oxygen in a bacterial system is linked to the speed of the impellors, such that the speed may be automatically controlled to try to raise the $DO_2$ by creating smaller bubbles via faster mixing before a greater rate of gassing is applied. This is vital in most systems as there is a limit to how much gas can be supplied to a system. Foam may occur if too much gas is added to a fermentation, which leads potentially to blocked filters as medium gets past the condenser and on to the exhaust filter. The other reason that the gassing rate is limited to a maximum of 2 vessel volumes per minute is that beyond this point cavitation may occur; this is when the impellor stops breaking up the gas bubbles and instead the impellor is surrounded by a gas pocket and any new gas is simply diverted around the impellors.

Combinations of impellors may be utilised such that a Scoping marine impellor may be used to create a downward force while a Rushton is used at the liquid gas interface to reduce foaming but also to create a larger surface area over which gas can be pulled into the system.

Many other impellor types do exist, for example, the Lightning range. Regardless of type and design, the requirement for good mixing without the creation of zones is the key consideration. The power that turns the impellor must be relayed from the motor to the impellor via a drive shaft.

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# MASS TRANSFER IN BIOREACTORS

Mass transfer occurs in mixtures containing local concentration variables.

## MT PLAYS A VITAL ROLE IN MANY REACTION SYSTEMS

For example, the mass transfer of Oxygen from the bubbles in the fermentation broth is slow, then the rate of cell metabolism can become dependent on the rate of oxygen supply from the gas phase.

## SO WHAT DRIVES MT?

Molecular diffusion is the movement of molecules of a given substance from an area of high concentration of that given compound to one of lower concentration. The term concentration gradient is used to describe this difference in concentration and molecules will move in a direction which will eventually destroy the gradient.

## WHAT HAPPENS IN GAS-LIQUID MASS TRANSFER

GLMT is unusual as molecules have to move between different phases (i.e. the gas phase to the liquid phase) across a boundary or interface.

*Film theory* attempts to explain the process of diffusion of Oxygen from the bubbles in the broth to the bulk liquid phase.

Let us consider the mass transfer of Oxygen from gas to liquid across the bubble interface. Oxygen is in much higher concentration in the bubble (21% of the volume of the air is Oxygen) also Oxygen is sparingly soluble in aqueous solutions. Therefore, a considerable driving force exists to move $O_2$ from gas to liquid is indicated in the fig. 1.

According to the film theory, the turbulence in each fluid dies away as you approach the interface. A thin film of stagnant fluid lies on each side of the interface. The only way for the Oxygen is molecular diffusion (normally in gasses or liquids we have convective mass transfer)

The concentration of any compound at the interface differs across the interface, even though the fluids at either side of the interface are supposed to be in equilibrium. This is simply due to the molecules relative solubility in each phase. As you move away from the interface, in either direction, the concentration of the molecule returns to the bulk liquid concentration that is regulated through convective mass transfer.

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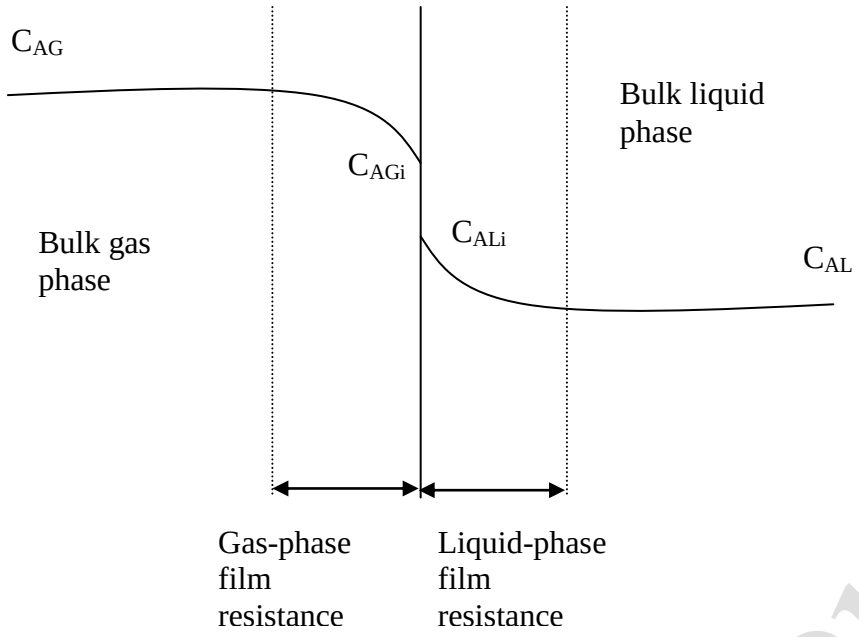



Figure 1: Film theory diagram reproduced from Doran

## CONVECTIVE MASS TRANSFER

**Definition:** Mass transfer occurring under the presence of bulk fluid motion.

Rate of MT is directly proportional to the driving force for transfer, and the area available for the transfer process.

Transfer Rate $\propto$ (transfer area) x (driving force).

The constant of proportionality is termed the Mass Transfer Coefficient.

Looking at film theory, the driving force for mass transfer from the bulk gas to the interface can be expressed as.

$$N_{AG} = k_G a(C_{AG} - C_{AGi}) \quad \text{---------- (1)}$$

and the rate of transfer through the gas liquid boundary as,

$$N_{AG} = k_L a(C_{ALi} - C_{AL}) \quad \text{---------(2)}$$

where $k_G$ is the gas phase MT coefficient and $k_L$ is the liquid phase MT coefficient. "a" is the interfacial area available for MT.

It is important to eliminate $C_{ALi}$ and $C_{AGi}$ as they are impossible to measure.

The first assumption we can make is that the interfacial gas concentration of the component in question is a linear function of the interfacial liquid concentration of the component. i.e.

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$$C_{ALi} = \frac{C_{AGi}}{m} \qquad \text{----------(3)}$$

where m is the distribution factor.

At steady state, there can be no accumulation of the component at the interface, i.e. anything transported from the gas phase to the interface is subsequently moved from the interface into the bulk liquid.

$$N_{AG} = N_{AL} = N_A \qquad \text{----------(4)}$$

By manipulating the previous four equations, it is possible to eliminate all interface terms in the relationship

$$N_A\left(\frac{1}{k_L a} + \frac{1}{m.k_G a}\right) = \frac{C_{AG}}{m} - C_{AL} \qquad \text{----------(5)}$$

The overall liquid phase mass-transfer coefficient, $K_L$ is defined as:

$$\frac{1}{K_L a} = \left(\frac{1}{k_L a} + \frac{1}{m.k_G a}\right) \qquad \text{----------(6)}$$

Therefore, we can simplify an expression to describe the transfer of a component from the gas phase to the liquid phase to:

$$N_A = K_L a\left(\frac{C_{AG}}{m} - C_{AL}\right) \qquad \text{----------(7)}$$

$\frac{C_{AG}}{m}$ is often referred to as $C^*_{AL}$, the concentration of component at the gas liquid interface that is in equilibrium with the gas phase concentration. A final representation of (7) can therefore be;

$$N_A = K_L a\left(C^*_{AL} - C_{AL}\right) \qquad \text{----------(8)}$$

In the case of Oxygen transfer, the key element of this course, Oxygen is sparing soluble in aqueous solutions, greater resistance to mass transfer is provided by the liquid film over the gas film. $k_G a$ is going to be much larger than $k_L a$, therefore from (6),

$$k_L a \approx K_L a \qquad \text{----------(9)}$$

and

$$N_A = k_L a\left(C^*_{AL} - C_{AL}\right) \qquad \text{----------(9.1)}$$

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It is important to note that $C^*_{AL}$ is the maximum attainable concentration of Oxygen in the liquid phase. Let us see now what impacts on this value.

## FACTORS THAT IMPACT ON $C^*_{AL}$

### 1. Gas Pressure and Oxygen Partial Pressure in the gas

The equilibrium relationship between oxygen partial pressure in the gas phase (and therefore total air pressure) and the liquid saturation oxygen concentration can be described by the following relationship.

$$p_{AG} = p_T \cdot \gamma_{AG} = HC^*_{AL} \quad \text{----------(10)}$$

What this equation means, is that the partial pressure of oxygen in the gas phase, $p_{AG}$, which is equivalent to the total pressure of the gas phase by the mole fraction of oxygen in the gas, $p_T \cdot \gamma_{AG}$, is equal to a constant times (H) the saturation concentration of oxygen in the liquid phase. This constant is known as Henry's constant (which is temperature dependant).

Obviously by increasing the partial pressure of Oxygen, either through enrichment or by general pressure increase will lead to a higher saturation concentration for oxygen in the liquid phase.

### 2. Temperature

Increasing temperature causes $C^*_{AL}$ to drop, so that the driving force for mass transfer is reduced.

**$O_2$ solubility and Henry's Law constant for pure water**

| ***Temperature*** ***($^oC$)*** | ***Saturation $O_2$ Conc.*** ***(mg/L)*** | ***Henry's Constant*** ***(atm L $mg^{-1}$)*** |
|---|---|---|
| *25* | *8.10* | *0.0258* |
| *35* | *6.99* | *0.0299* |

Above 40°C the solubility of Oxygen drops significantly.

### 3. Solutes

Fermentation media typically contain significant concentrations of salts, sugars and other dissolved components. The presence of such solutes has a significant impact on the solubility of Oxygen at a given temperature and pressure.

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In general, the solubility of Oxygen decreases with increasing concentration of solutes. The saturation concentration of oxygen can potentially decline by 40-50% in typical fermentation media over that of deionised water in the same environmental conditions.

## FACTORS THAT IMPACT ON $K_LA$

### Antifoams

Most fermentations end up producing agents that produce and stabilise foam, such as proteins, polysaccharides and fatty acids.

Most fermentation engineers will deploy antifoam agents in anticipation of this problem. Antifoam agents, in general reduce the surface tension in the foam, generally causing the foam to collapse.

Reducing the surface tension, generally results in smaller bubbles. This may initially be seen as good for $k_La$ (the interfacial area is increased by having smaller bubbles).

However, the interface mobility of the bubbles is significantly decreased by the antifoam agent, resulting in a decrease in the $k_L$ element. The decrease in $k_L$ generally outweighs the increase in a, resulting in a net drop in the gas-liquid mass transfer coefficient. This can often be by up to a factor of 10.

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# RHEOLOGY OF FERMENTATION FLUIDS

Fluids may be described as Newtonian or non-newtonian depending on whether their rheology (flow) characteristics obey Newton's law of viscous flow. Consider a fluid contained between two parallel plates area *A* and distance *x* apart. If the lower plate is moved in one direction at a constant velocity, the fluid adjacent to the moving plate will move in the same direction and impart some of its momentum to the 'layer' of liquid directly above it causing it, also, to move in the same direction at a slightly lower velocity. Newton's law of viscous flow states that the force, *F,* opposing motion at the interface between two liquid layers, flowing with a velocity gradient *dv/dx,* is given by the equation:

$$F = \mu A(\mathrm{d}v/\mathrm{d}x) \quad \text{----------(1)}$$

where μ *is* the fluid viscosity, which may be considered as the resistance of the fluid to flow.

Equation (1) may be written as:

$$F/A = \mu(\mathrm{d}v/\mathrm{d}x)$$

*F*/ *A* is termed the shear stress $(\tau)$ and is the applied force per unit area; $du/dx$ is termed the shear rate $(\gamma)$ and is the velocity gradient. Thus:

$$\tau = \mu\gamma \quad \text{----------(2)}$$

Equation (2) conforms to the general relationship:

$$\tau = K\gamma^n \quad \text{----------(3)}$$

where, *K* is the consistency coefficient and *n* is the flow behaviour index or power law index.

For a Newtonian fluid *n* is 1 and the consistency coefficient is the viscosity which is the ratio of shear stress to shear rate. Thus, a plot of shear stress against shear rate, for a Newtonian fluid, would produce a straight line, the slope of which would equal the viscosity. Such a plot is termed a rheogram shown in Fig.1

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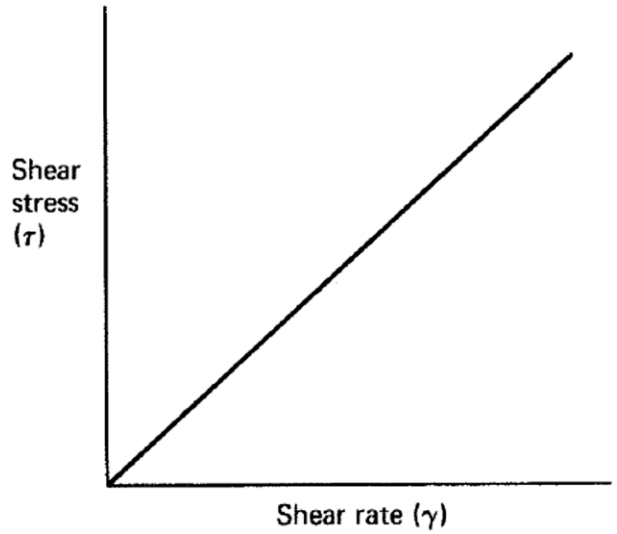



Figure.1. A rheogram of a Newtonian fluid.

Thus, a Newtonian liquid has a constant viscosity regardless of shear, so that the viscosity of a Newtonian fermentation broth will not vary with agitation rate.

However, a non-Newtonian liquid does not obey Newton's law of viscous flow and does not have a constant viscosity. The value for *n* of such a fluid deviates from 1 and its behaviour is said to follow a power law model. Thus, the viscosity of a non-Newtonian fermentation broth will vary with agitation rate and is described as an apparent viscosity $(\mu_a)$. A plot of shear stress against shear rate for a non-Newtonian liquid will deviate from the relationship depicted in Fig.2, depending on the nature of the liquid. Several types of non-Newtonian liquids are recognized.

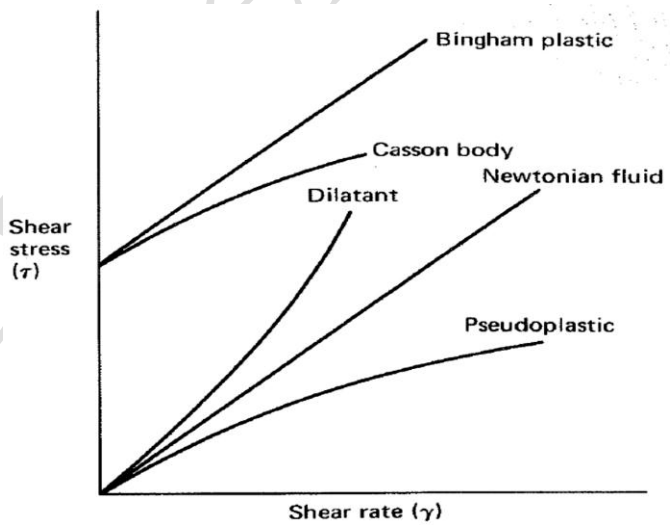



Figure 2. Rheogram of fluids of different properties.

## BINGHAM PLASTIC RHEOLOGY

Bingham plastics are similar to Newtonian liquids apart from the fact that shear rate will not increase until a threshold shear stress is exceeded. The threshold shear stress is termed the yield stress or yield value $\tau_o$. A linear relationship of shear stress to shear rate is given once the yield stress is exceeded and the slope of this line is termed the coefficient of rigidity or the plastic viscosity. Thus, the flow of a Bingham plastic is described by the equation:

$$\tau = \tau_0 + n\gamma$$

where, *n* is the coefficient of rigidity and $\tau_o$ is the yield stress. There have been some claims of mycelial fermentation broths displaying Bingham plastic characteristics. Everyday examples of these fluids include toothpaste and clay.

## PSEUDOPLASTIC RHEOLOGY

The apparent viscosity of a pseudoplastic liquid decreases with increasing shear rate. Most polymer solutions behave as pseudoplastics. The decrease in apparent viscosity is explained by the long chain molecules tending to align with each other at high shear rates resulting in easier flow. The flow of a pseudoplastic liquid may be described by the power law model, equation

$$\tau = K(\gamma)^n$$

*K* has the same units as viscosity and may be taken as the apparent viscosity. The flow-behaviour index is less than unity for a pseudoplastic liquid, the smaller the value of n, the greater the flow characteristics of the liquid deviate from those of a Newtonian fluid. Equation in the logarithmic form can be written as:

$$\log\tau = \log K + n\log\gamma$$

Thus, a plot of log shear stress against log shear rate will produce a straight line, the slope of which will equal the flow-behaviour index and the intercept on the shear stress axis will be equal to the logarithm of the consistency coefficient. Many workers have demonstrated that mycelial fermentation broths display pseudoplastic properties.

## DILATANT RHEOLOGY

The apparent viscosity of a dilatant liquid increases with increasing shear rate. The flow of a dilatant liquid may also be described by equation

$$\tau = K\gamma^n$$

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In this case the value of the flow-behaviour index is greater than 1, the greater the value the greater the flow characteristics deviate from those of a Newtonian fluid. Thus, the values of $K$ and $n$ may be obtained from a plot of log shear stress against log shear rate. Fortunately this type of behaviour is not exhibited by fermentation broths an everyday example is liquid cement slurry

## CASSON BODY RHEOLOGY

Casson (1959) described a type of non-Newtonian fluid, termed a Casson body, which behaved as a pseudoplastic in that the apparent viscosity decreased with increasing shear rate but displayed a yield stress and, therefore, also resembled a Bingham plastic. The flow characteristics of a Casson body may be described by the following equation:

$$\sqrt{\tau} = \sqrt{\tau_0} + K_c \sqrt{\gamma}$$

where, $K_c$ is the Casson viscosity. Therefore, to determine the rheological nature of a fluid it is necessary to construct a rheogram which requires the use of a viscometer which is accurate over a wide range of shear rates.

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# SCALE-UP CONCEPTS

Steps involved in scale up of a fermenter:

1) Choose fermenter volume required based on desired capacity.

2) Algorithm for fermenter

$$V = \frac{v_0 C_p}{Y_{p/e} Y_g}$$

3) Choose impeller diameter, Di.

4) Calculate reactor dimension (e.g., $D_T$ = tank diameter) based on geometric similarity, with the impeller diameter being the characteristic length, for example,

$$\frac{D_{T\,\mathrm{PLANT}}}{D_{T\,\mathrm{LAB}}} = \frac{D_{i\,\mathrm{PLANT}}}{D_{i\,\mathrm{LAB}}}$$

Alternately, we could have chosen the tank diameter, $D_T$, in step 2 and then used Equation (A) to calculate the impeller diameter, $D_i$.

5) Calculate impeller speed, $N$.

$$(\pi D_i N)_{\mathrm{PLANT}} = (\pi D_i N)_{\mathrm{AB}}$$

Then

$$N_{\mathrm{PLANT}} = N_{\mathrm{LAB}} \frac{D_{\mathrm{LAB}}}{D_{\mathrm{PLANT}}}$$

6) Choose mass transfer correlation for $k_b a_b$.

7) Calculate gas flow rate using the correlation and setting

$$(k_b a_b)_{\mathrm{PLANT}} = (k_b a_b)_{\mathrm{LAB}}$$

the mass transfer coefficient $k_L a_b$ depends on the following equipment parameters

$$k_b a_b \sim \frac{1}{D_i^2} (D_i^2 N)^{1.1} \left(\frac{4Q}{\pi D_i^2}\right)^{0.45} \sim D_i^{0.2} N^{1.1} \left(\frac{Q}{D_i^2}\right)^{0.4}$$

Then

$$Q_{\mathrm{PLANT}} = Q_{\mathrm{LAB}} \frac{D_{i\,\mathrm{PLANT}}^2}{D_{i\,\mathrm{LAB}}^2} \left[\left(\frac{D_{i\,\mathrm{LAB}}}{D_{i\,\mathrm{PLANT}}}\right)^{0.2} \left(\frac{N_{\mathrm{LAB}}}{N_{\mathrm{PLANT}}}\right)^{1.1}\right]^{1/0.45}$$

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8) Calculate power requirement.

$$P(w) = p_0 \rho \left( \frac{RPM}{60} \right)^3 D^5$$

$p_o$ is the power number of the impeller. For a fully immersed Rushton impeller it is 5, and if it is very near the liquid surface it is less than 5. ρ is density (kg/m$^3$). *RPM* are revolutions per minute.

- *D* is the diameter of the impeller. A typical correlation for $k_La$(in $hr^{-1}$):

$$k_L a = 0.1 \left( \frac{P(w)}{M} \right)^{0.4} \upsilon_{gs}^{0.6}$$

- *M* is the mass of the fluid.
- *0.1* is experimentally determined (should be determined for each culture broth).
- $\upsilon_{gs}$ is superficial gas velocity

Recall that:

$$\boxed{OUR = k_L a\left(C^* - C_L\right)}$$

*OUR* is oxygen uptake rate.
$C^*$ is the sat'n concentration of $O_2$.
$C_L$ is the actual DO.

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# DESIGN OF FERMENTATION MEDIA

## MEDIA TYPES

Media can be solid, semi-solid or liquid. For the purposes of this chapter solid and semisolid media will not be considered, only liquid media used in bioprocessing will be covered. Most liquid media can of course, readily be converted to solid media by addition of a suitable gelling agent, such as agar or gellan. Just make sure the gelling agent is fully dissolved and dispersed before attempting to sterilize the medium.

## MICROBIOLOGICAL CULTURE MEDIA

Media used in bioprocessing can be broadly divided into three categories based on their composition: synthetic, semi-synthetic and complex.

## SYNTHETIC MEDIA

Synthetic media is fully chemically defined. All the components is known, as the specific concentration of each of the components. Such media are usually quite simple, containing a carbon/energy source, a nitrogen source, and a range of salts (usually inorganic). However, if the cell line chosen is particularly fastidious and/or has specific growth requirements, the complexity of the synthetic medium may increase significantly, e.g. in the drive to replace fetal calf serum for cultivation of animal cells the synthetic media derived have between 70 and 80 different components, depending on the cell type grown.

Synthetic media are useful in research and laboratory situations where experimental accuracy is paramount and data interpretation needs to be clear. In general, such media are much more costly than other media types, especially if specific components such as vitamins and growth factors are required, as these ingredients tend to be very expensive when supplied in the pure form. Yields of cells tend to be much lower than those obtained when the same cell line is grown in either semi-synthetic or complex media, as the synthetic media often do not match an organism's or cell line's exact requirements. It is, of course, easier to investigate the impact of single nutrients upon cell growth and physiology using these media than the more complex ones described below. Likewise, since medium composition can be defined and delivered in a reproducible fashion, in principle, the fermentations carried out using such formulations should exhibit less variability. Such media are generally fully soluble and provide the fewest downstream processing challenges in an industry context or purification/recovery challenges in a small laboratory context.

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## SEMI-SYNTHETIC MEDIA

Semi-synthetic media is largely chemically defined as above, but include one or more poorly specified component(s) of variable but controlled composition, for example, yeast extract, which is particularly useful if a cell line requires a range of B vitamins. Such media are useful in research and laboratory situations where a particular organism has a requirement for a substrate that would be too expensive to supply in the pure form on a routine basis. Such components are also valuable where the specific nutritional needs are not well known, and a rich source of many nutrients such as yeast extract is used in a 'shotgun' approach to provision of minor nutrients.

Plant, animal, fish and microbial extracts have routinely been used in the past to supply vitamins and essential growth factors for specific organisms. The current trend in medium design is to avoid extracts of animal origin, especially bovine sources, as there is potential health risks associated with these products. There is thus a void in the market that is waiting to be filled by a substrate that is cheap, reliable and contains a range of vitamins and growth factors.

It is wise to remember that, when using media of this nature; batch-to-batch variability of the poorly specified component is a very real possibility. Where accuracy is paramount it may be necessary to analyse the substrate to define the specific component that the material is supplying. This obviously has a cost implication, as routine assays have both consumable and labour costs associated and, perhaps more importantly, time needs to be factored in to carry out these assays. Such control of the quality of raw materials is standard practice at industrial scale, but has time and equipment resource implications. Given that variation between batches of these raw materials exists, and can affect the fermentation progress, how can the impact of this be minimised in a small laboratory without the time and resources to monitor 'quality' of raw materials  Simple batch mixing is one route to keep this to a minimum.

Use of semi-synthetic media can also impact on the routine analyses required in monitoring the process. For example, using a substrate such as yeast extract to supply vitamins to the cells would mean that a routine, relatively simple, assay for quantifying the nitrogen in the medium (e.g., $NH_3$ content using an ammonia electrode) becomes a lot more complex as the extract will also contain complexed nitrogen sources. The samples would therefore require additional assays to be carried out (in this case a Kjeldahl digestion assay) to determine the total nitrogen available to the cells. Yeast extract also adds some fermentable carbohydrates to the medium, again complicating the clear identification of C substrate usage in these media.

Critical evaluation of the medium to be used should therefore balance the gain achieved in using undefined components against the disadvantages associated with the more complex analyses. It may after all, be more cost effective to use the expensive pure compound.

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## COMPLEX MEDIA

Complex media are largely composed of substances that are usually of plant or animal origin, and that have ill-defined and variable composition. These materials vary from batch to batch, and composition is influenced by time of year, location of origin, and small changes in production methods. Seasonal availability may change appreciably. They are usually relatively cheap, can be supplied in bulk and the source should be reliable. Substrates such as beet and cane molasses, corn steep liquor, soya bean meals and extracts, whey powders, hydrolysed starches and a range of many different oils all fall into this category. Complex media are used for many biotechnological processes, especially those that produce high volume, low value products, e.g. single cell protein (bakers' yeast, Quorn). These complex substrates will normally meet more than one requirement of the cell, perhaps supplying carbon and nitrogen, e.g. rapeseed (canola) oil will supply both carbon and nitrogen to the process, and many of the other complex substrates will supply vitamins, growth factors, and many elemental needs of the cell in addition to being a source of carbon. Many industrial antibiotic processes still employ varying amounts of oils such as rapeseed, as in addition to the nutrients added, the oil may specifically enhance antibiotic synthesis, and also help to suppress foam formation. Foam formation tends to be far worse in complex and semi-synthetic media than in fully synthetic, due to the presence of amino acids, and proteinaceous material in the former. This greater need for antifoam must be anticipated with these medium types. At research level it is rare to use such substrates, unless an industrial process is to be mimicked exactly in the research laboratory, as each one may contain a wide range of impurities and the composition of the substrate can be highly variable depending on what part of the world the substrate is grown in, soil type, climate, etc. If the use of such materials is unavoidable at small scale, it implies high in-process variability, and even with careful planning of experimentation (e.g., using design of experiment software packages), you should expect that more replicate fermentations will be required to obtain clear results. For example, the composition of molasses varies according to whether the molasses is obtained from beet or cane sources.

Routine use of such a substrate implies routine analyses of every batch used. Molasses may also contain a high concentration of iron, as well as other elements, and iron can be toxic to many cells. The substrate may therefore be subject to ion-exchange chromatography before it can be used, and thus incurs capital and labour costs. Again the critical evaluation needs to balance the role and advantages of the complex medium against the disadvantages. When dealing with complex media it is useful to consider what is scheduled to happen downstream of the bioreactor, as the physical characteristics of the substrate can impact greatly on factors such as filtration, centrifugation and purification stages.

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# MEDIUM COMPONENTS

The starting point, when considering which medium components to supply to a cell, is a consideration of the elemental balance of the cell. This may not necessarily be readily available in the literature, often being considered too routine to be worthy of publication. A well washed, dried sample of the cells can be analysed by any well found chemistry laboratory. Cells grown for this purpose should not be grown on a minimal synthetic medium as the cells may not grow to their full potential on this type of medium. Instead cells should be grown on a complex medium with the potential to supply most of the cells' requirements. The cells must be thoroughly washed in an osmotically favourable solution to ensure that any residual sugars, etc., are removed from the cell surface and dried under vacuum before sending for analysis.

It can be seen that while the carbon content of the different groups is very similar, the nitrogen content varies considerably. This is a reflection of the way in which the different organisms grow. Fungi, having large areas of the hyphae that are empty and no longer involved in biochemical activities, and thus contain no nucleic acids or proteins, have considerably less nitrogen than yeasts or bacteria. In addition, different stages of culture growth and different specific growth rates may produce a cell composition that varies significantly. A good example is seen in chemostat cultures of the yeast (*Saccharomyces cerevisiae*) where cells grown at low dilution rates may typically have 50% more protein content than those produced at high dilution rates.

## CARBON SOURCE

A carbon source is necessary to provide the cell with energy as well as the material with which to grow and synthesize arrange of primary and secondary metabolites. The best energy source depends on the type of organism utilised, e.g. autotroph, chemotroph, etc.

There is obviously a wide range of carbon sources and the one chosen should be appropriate to the organism but also to the economics of the process. At research scale the latter tends to be less important, but it should be borne in mind if the objective of the programme is to develop an industrially relevant process. This section will look at several carbon sources and indicate the advantages and disadvantages of each. A typical microorganism is approximately 50% carbon, making carbon the most significant substrate and care should be taken to ensure that the carbon concentration does not become limiting. The yield of biomass on carbon is approximately 0.5, which means that if a biomass concentration of 50 g/L is required, 100 g/L of the carbon source must be supplied, albeit not necessarily all at once if fed-batch culture is being utilised.

The great majority of laboratory and industrial fermentations tend to use a very limited range of easily utilizable substrates to supply the energy and Carbon requirements of cultures. This does not imply that other Carbon and energy sources could not be used, just that this limited range is usually available and methods for preparing and analyzing the consumption rates of such substrates are well understood. The production of pharmaceuticals, especially biopharmaceuticals by fermentation, takes place in the context of an industry that is, with the

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possible exception of the nuclear industry, perhaps the most regulated in the world. This fact accounts for a degree of 'conservatism' in terms of nutrient supply.

## GLUCOSE

Glucose is universally acceptable for growth of most cell lines, be they animal, plant or microbial. Supplied as a powder in the pure form, the substrate is readily available, reliable, easily stored, easily handled and has no significant implications for health and safety. These qualities make glucose a popular choice of carbon source.

There some drawbacks to using glucose, notably:

- The possibility of the organism suffering from the 'Crabtree effect' if glucose is over supplied in the initial stages of growth;
- The loss of available substrate to the Maillard reaction, which can occur if glucose is sterilized with a nitrogen source present.

Both of these situations can be overcome readily, by careful monitoring and control of glucose feed in the first case and by either sterilizing the substrates separately or by using filtration sterilization methods in the second.

A popular method of supplying glucose on a larger scale is to use glucose syrups, manufactured by the hydrolysis of starch (a substrate which itself tends to be insoluble and consequently unavailable to the cells, difficult to handle, but is readily hydrolysed to glucose). The drawbacks remain the same, with the additional complications of transportation, storage and handling of a liquid. Formulation of the medium and delivery to the bioreactor may also be considerations that have to be taken into account. The main advantage to using glucose syrups is cost, as it is significantly cheaper than powdered glucose.

## SUCROSE

Often the sugar of choice in research laboratories, sucrose, is utilised by many but not all cell lines. Commercially available sucrose comes in many forms and grades, from pure granulated forms to complex molasses solutions. Sucrose does not tend to suffer from the same drawbacks as glucose and, although it can be subject to caramelisation when over sterilized, generally it can be autoclaved/sterilized with nitrogen compounds without the same problems as glucose. If using molasses, sufficient cold storage must be available to handle the needs of the laboratory as the range of different substrate available means that the molasses can be readily contaminated, and although sterilization can destroy any contaminating organisms, the amount of available sucrose to the organism of choice will decrease. Again mixing of molasses from several batches can help iron out some variability contributed by this source.

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## LACTOSE

Lactose can only be utilized by a few cell types, e.g. Escherichia coli, and is usually only metabolized very slowly. The sugar is only useful for some commercial processes that use the complex substrate whey, a by-product of the dairy industry, which contains both lactose (approximately 50 g lactose per liter of whey and 4% w/w nitrogen). Available as a liquid or as whey powder there are choices available regarding storage and handling. Some fermentation media in the past typically batched in both glucose, to achieve rapid culture growth, and lactose, which was only utilised after glucose exhaustion, and the slow rate of usage fuelled, e.g. antibiotic synthesis. Controlled feeding of glucose usually obviates the need for this nowadays.

## OTHER SUGARS

Other sugars can be used as substrates but tend to be very specific to the cell line chosen, e.g. melobiose is sometimes used for the cultivation of yeast cells. Sugars falling into this category tend to be too expensive to use on a routine basis. If choosing an unusual sugar the questions to be posed are 'Why is this sugar required?' and 'What advantage will the organism gain?'

## OILS

A number of oils are now routinely used as carbon sources in the bioprocess industries. Rapeseed oil (canola oil in the USA/Canada) and methyl oleate are two popular choices. Representing both a carbon and an energy source, oils are economical and readily available. The plants grown to produce the oils will grow in many climatic zones and are readily cultivated, so supplies will be reliable. Variations do exist in the composition/ quality of the oils and they must be regularly assayed to check for variation in component content between batches. In a laboratory situation care must be taken to ensure that suitable storage facilities (4 °C) are available.

On a small scale delivery of the oil can be an issue. If operating a batch bioreactor, the oil can be added at the start, before sterilization and appropriate mixing within the reactor will disperse the oil throughout the vessel. However many small scale reactors are poorly mixed and there may be difficulties that ensue as a consequence of this. Before selecting oil as a suitable substrate, the mixing in the bioreactor should be checked to make sure, it is suitable (particularly at 1–2 liter scale). If using fed batch culture at laboratory scale, calculation of the amount of oil required per unit volume per minute must be very accurate and should take into account all substrates in the oil.

Flow rates need to be checked to ensure that sufficient oil is delivered sometimes it can be very difficult to achieve if the correct pumps for substrate addition are not chosen. Low flow rates often involve working at the lowest speed of the pump and this is usually where pumps are most inaccurate. Choosing the correct type of low speed pump is essential it is not necessarily going to be the pump supplied by the bioreactor manufacturer, who tends to consider only non-viscous

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fluids such as routine feeds and pH titrants. To complicate the issue even further the viscous nature of the oil can make additions difficult and often inaccurate.

To maximize accuracy, the following should be observed:

- Know the exact substrate concentrations in the oil
- Try to use a combination of tubing bore (internal diameter), which lets the pump be operated in the middle of its range/speed – this may involve purchase of a special pump fit for purpose – the internal and external diameter of the tubing round the pump head can always be altered to help improve control
- Regularly check the substrate reservoir and calculate the delivery rate. With slow speed pumps, delivering a small volume per hour this can often be achieved more accurately by taking a reading every 4 hours rather than every hour.

An additional benefit of using oil is that it can be used as an antifoaming agent, thus avoiding the use of non-metabolisable antifoams that can potentially interfere with downstream processing activities. The oil is broken down as it is utilised by the cells and thus does not cause problems later. A few drops of oil can be sufficient to prevent foaming, particularly when dealing with a stirred, aerated proteinaceous broth. If using oil for this purpose it is advisable to know exactly how much of the oil is used, so that the carbon and nitrogen contribution to the medium is known. Sometimes this knowledge can only be gained from experience of the process.

Oil is often filtered/sterilized at the large scale for the sake of simplicity, and increasingly this is a convenient method for small scale processes too.

## NITROGEN SOURCES

The amount of nitrogen in any medium really determines the amount of biomass that will be achieved for the particular cell line, given that plenty of carbon is available and all required nutrients are present in the initial medium. Nitrogen is required for growth and synthesis of, for instance, proteins and nucleic acids. The type and sources are variable. We have already seen that some nitrogen can be delivered as a side benefit to the addition of complex nutrient sources.

Points to consider before selecting the nitrogen source are as follows:

- What do we want the process to do, manufacture cells (i.e., biomass), intracellular or extracellular product, or both .In many fermentation processes growth alone is not the target, the synthesis of a specific product is required.
- Do we want the nitrogen to run out at any point in the process? For example, if dealing with a true secondary metabolite, such as an antibiotic, it is necessary to design the medium so that one substrate becomes limiting in order that growth ceases or slows down considerably to allow production of the product.

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This is often achieved by allowing nitrogen to become limiting once sufficient biomass has been achieved this cannot be the case if a protein is the required product for obvious reasons.

How will pH control be achieved? In a well controlled bioreactor this is not an issue, but if a buffered system is being employed it is worthwhile considering what contributions salts in the medium may make to the overall pH of the broth. In some instances, the culture may be fed N source in the form of $NH_3$ or $NH_4OH$ at a rate just matching its metabolic activity. This also achieves pH control in acid-forming fermentations.

A traditional rule-of-thumb guide to a balanced growth medium suggests an approximate ratio of nutrients required of carbon : nitrogen : phosphorus of 100 : 10 : 1. Therefore, for every 10 $gL^{-1}$ carbon source, 1 $gL^{-1}$ nitrogen is required. A common error here is to assume that 1 $gL^{-1}$ $(NH_4)_2SO_4$ contains 1 $gL^{-1}$ N, which it obviously does not, so remember to take into account the other elements that may be in the chosen nitrogen source. Likewise, this is important when assessing the effects of different N providing salts: make sure the absolute amounts of N being provided are equal otherwise what is observed will be a function of N availability not necessarily the identity of the N source. Typical nitrogen sources are discussed in the following sections.

## AMMONIA

Often used in industrial processes, ammonia is not a common source of nitrogen used in research laboratories because of the volatility of the liquid and the associated handling problems. Health and safety issues are a real risk with this nitrogen source, and considerable thought is needed if this is to be the substrate of choice. Special handling equipment, storage reservoirs and breathing equipment are all required, as are additional safety precautions should there be a leak.

The advantages of using ammonia are that it readily dissolves in the medium, it is immediately available to the cell, and it is cheap (although storage and handling equipment may not be).

## NITROGEN BASED SALTS

Ammonium sulfate, $(NH_4)_2SO_4$, and ammonium chloride, $NH_4Cl$, are common nitrogen sources found in the literature. Ammonium is very utilizable by many organisms and use of a salt is a cheap and convenient method of supplying the cell with what it requires. In addition, it is easy to calculate the exact amount of nitrogen being supplied to the cells. Storage of the substrate is easy, requires no special storage areas or equipment, the salts just needing to be stored in a cool dry place. Analysis of the nitrogen content of the resultant fermentation broth is usually very simple and can be achieved using a simple assay and an ammonia probe.

It is useful to consider other requirements the organism may have, for example if the cell has a requirement for sulfur, then ammonium sulfate would be the salt to choose, so that one salt satisfies more than one requirement. This is not only economical it also makes formulation of the

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medium much simpler. As a rule, try to keep the medium formula as simple as possible to minimize the chance of errors and to reduce medium generated variability.

A word of warning when dealing with any salts required in bioprocess media is to watch the combination of salts used, and the order in which they are added. An incorrect combination can lead to a batching vessel full of salt crystals. Be systematic and make sure the order of addition of ingredients is consistently adhered to.

## COMPLEX NITROGEN SOURCES

There is a number of excellent complex nitrogen sources available, e.g. yeast extract, soya bean meal (8% w/w nitrogen), and corn steep liquor (4% w/w). All supply a significant amount of nitrogen, in addition to other essential nutrients such as carbon, vitamins, and minerals.

As with the complex carbon sources, there is significant batch-to-batch variability and every batch of the substrate has to be analysed for component composition before use. Analysis of the fermentation broth is also more complicated and often involves acid hydrolysis of the broth in order to get the nitrogen into a readily assayable form. This involves considerable time, labour and extra cost.

## OTHER SUBSTRATES

### ELEMENTS

There are several elements that are essential from the growth of microorganisms. Each must be provided, generally in the form of an inorganic salt, in order for the cell to grow and function properly.

- **Calcium.** This is required to stabilise cell walls, and is particularly important if the cells form endospores (which contain a high concentration of calcium dipicolinate).
- **Magnesium.** This often acts as a cofactor for the activity of enzymes, can play a significant role in membrane structure and function.
- **Phosphorus.** This is required for the production of phospholipids, nucleic acids and the generation of energy (ATP, ADP).
- **Potassium.** This is involved in many reactions within cells and is required by all cells, animal or microbial.
- **Sodium.** A few species have an absolute requirement for sodium, e.g. marine species and halophiles. Many cells however, have no obligate requirement for this element.
- **Sulfur.** This required for amino acid synthesis (cysteine, cystine and methionine), for the production of some vitamins, e.g. biotin, and for coenzyme function in many, but not all, cells.

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Again, if complex media are being utilised, there may be a reduced or even no need to supply these elements separately. In defined media it is often necessary to add the elements if the cell line is to grow and function normally.

## TRACE ELEMENTS

Trace elements are regarded as micronutrients, being required only in tiny amounts, usually µg quantities per litre. The functions of trace elements within the cell are many and varied but are usually associated with enzyme activity, where the trace element forms part of the enzyme or functions as a catalyst for the biochemical reaction the enzyme is involved with. Again, where complex or semi-synthetic media are utilised, it may not be necessary to supply these elements separately. Local water supplies may even contain adequate amounts of these elements. If, however, a very pure synthetic medium utilized in the laboratory, or if a cell has an absolute requirement for a particular element or elements, they must be added to the medium. Very high cell density fed-batch cultures (e.g., Pichia pastoris for synthesis of recombinant proteins) can run out of trace elements unless care is taken to feed these at rates matching feeds of macronutrients. Think ahead, and do not simply assume that there will be such surplus of these elements batched in at the start that it obviates the need to add these during the process.

Typical trace elements include iron (Fe, probably one of the more important elements considering its role in cellular respiration), copper (Cu), cobalt (Co), manganese (Mn), molybdenum (Mo), chromium (Cr, animal cells only) to name only a few. Analysis of the literature should reveal which trace element or elements are required by the organism or cell line chosen.

It is usual to make up a trace element solution containing all the elements that are required by the cell, and to add a few millilitres of the resultant solution to the medium. This can readily be carried out before sterilization of the bulk medium. Stored carefully, the trace element solution can last many weeks but should evidence of precipitation be seen, e.g. one can often see the iron salts starting to coat the container, a new solution should be made up. The caveats above regarding order of addition in batching are also applicable here.

## GROWTH FACTORS

Growth factors, like trace elements, are required only in tiny quantities by the cell. Often these can be manufactured by the cell itself; however, some cells do not have the ability to manufacture certain key growth factors and they must be supplied to the cell in the medium. They are invariably organic compounds, such as vitamins, purines, pyrimidines, and amino acids. Such compounds are usually very expensive to add in the pure form and it is more normal in bioprocessing to use a cheaper version, which may not be as pure but that provides all the cell needs. If added in pre-form, they will usually not be sterilized using pressurized steam in an autoclave or in situ in an SIP fermenter, as they may well be heat labile.

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A very good example of such a substrate is yeast extract. Synthesized from yeast cells (many microorganisms have to ability to manufacture many growth factors) yeast extract contains a number of growth factors, the foremost and most important of which are the B vitamins. Vitamins are the most commonly used growth factors in cell culture media and supplying them in the form of yeast extract, or other plant and animal extracts, is cheap and convenient.

## INHIBITORS

Obviously, any medium formulation should be free of compounds that are known to inhibit the cell to be cultivated. However, most cell types, if overfed, will exhibit undesirable activities and may well excrete toxic by-products or waste products that can lead to slow growth or product synthesis, or termination of the fermentation process itself. In general, with synthetic or semi-synthetic media, known inhibitors are easily avoided. With complex media, however, there can be inhibitors present that have to be removed prior to use. For example, molasses contains a high concentration of iron, which can be toxic to the cell and this has to be removed (or the concentration significantly reduced) if molasses is to be used as the substrate.

Care also has to be taken where a substrate can adversely affect the cell line if present at too high a concentration. Glucose is a particular culprit here as desirable metabolic activities in many organisms are subject to repression by glucose, or excess glucose may simply cause 'overflow' metabolism to occur, e.g. acetic acid excretion by E.*coli*. If this is the case then glucose will usually have to be delivered in a fed-batch mode such that the concentration of residual glucose in the vessel is maintained below levels likely to cause repression or inhibition.

The best view of fermentation processes (especially fed-batch) is that of a balancing act, with the aim being to supply a medium of the right composition, at the right rate, in line with the oxygen transfer rate of the vessel to encourage the desired culture activity and to avoid or minimize side or waste product accumulation.

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# VARIOUS TYPES OF MICROBIAL AND ENZYME REACTORS

## MICROBIAL REACTORS

### THE WALDHOF-TYPE FERMENTER

The investigations on yeast growth in Sulphite waste liquor in Germany, Japan and the United States of America led to the development of the Waldhof-type fermenter have given a description of a production vessel based on a modification of the original design of Zellstofffabrik Waldhof. The fermenter was of carbon steel, clad in stainless steel, 7.9 m in diameter and 4.3m height with a centre draught tube 1.2 m in diameter. A draught tube was held by tie rods attached to the fermenter walls. The operating volume was 225,000 $dm^3$ of emulsion (broth and air) or 100,000 $dm^3$ of broth without air. Non-sterile air was introduced into the fermenter through a rotating pin-wheel type of aerator, composed of open-ended tubes rotating at 300 rpm. The broth passed down the draught tube from the outer compartment and reduced the foaming.

### THE TOWER FERMENTER

It is difficult to formulate a single definition which encompasses all the types of tower fermenter. Their main common feature appears to be their height: diameter ratio or aspect ratio. Such a definition has been given by Greenshields *et al.* (1971) who described a tower fermenter as an elongated non-mechanically stirred fermenter with an aspect ratio of at least 6:1 for the tubular section or 10:1 overall, through which there is a unidirectional flow of gases. Several different types of tower fermenter exist.

### CYLINDRO-CONICAL VESSELS

The use of cylindro-conical vessels in the brewing of lager was first proposed by Nathan (1930), but his ideas were not adopted for the brewing of lagers and beers until the 1960s (Hoggan, 1977). Breweries throughout the world have now adopted this method of brewing. These vessel consists of a stainless-steel vertical tube with a hemispherical top and a conical base with an included angle of approximately 70°.

Aspect ratios are usually 3:1 and fermenter heights are 10 to 20m. Operating volumes are chosen to suit the individual brewery requirements, but are often 150,000 to 200,000 $dm^3$. Vessels are not normally agitated unless particularly flocculent yeast is used, but small impellers may be used to ensure homogeneity when filling with wort (Boulton, 1991). In the vessel, the wort is pitched (inoculated) with yeast and the fermentation proceeds for 40 to 48 hours. Mixing is

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achieved by the generation of carbon dioxide bubbles that rise rapidly in the vessel. Temperature control is monitored by probes positioned at suitable points within the vessel. A number of cooling jackets are fitted to the vessel wall to regulate and cause flocculation and settling of the yeast Boulton, 1991). The fermentation is terminated by the circulation of chilled water via the cooling jackets which results in yeast flocculation. Thus, it is necessary to select a yeast strain which will flocculate readily in the period of chilling. Part of this yeast may be withdrawn and used for repitching another vessel. The partially cleared beer may be left to allow a secondary fermentation and conditioning.

Some of the advantages of this vessel in brewing are:

- Reduced process times may be achieved due to increased movement within the vessel.
- Primary fermentation and conditioning may be carried out in the same vessel.
- The sedimented yeast may be easily removed since yeast separation is good.
- The maturing time may be reduced by gas washing with carbon dioxide.

### THE DEEP-JET FERMENTER

Some designs of continuous culture fermenter achieve the necessary mechanical power input with a pump to circulate the liquid medium from the fermenter through a gas entrainer and back to the fermenter. Two basic construction principles have been used for the gas entrainer nozzles. The injector and the ejector. In an injector a jet of medium is surrounded by a jet of compressed air. The gas from the outlet enters the larger tube with a nozzle velocity of 5 to 100 $ms^{-1}$ and expands in the tube to form large air bubbles, which are dispersed by the shear of the water jet. In an ejector the liquid jet enters into a larger converging-diverging nozzle and entrains the gas around the jet. The gas which is sucked into the converging-diverging jet is dispersed in that zone. One of the industrial-scale fermenters using the ejector design is marketed by Vogelbusch (Vogelbusch AG, Vienna, Austria). Partially aerated medium is pumped by a multiphase pump through a broth cooler to an air entrainer above the fermenter. The air-medium mixture falls down a slightly conical shaft at a high velocity and creates turbulence in the fermenter. Two-thirds of the exhaust gas is vented from the fermenter headspace and the remainder via the multiphase pump. Oxygen-transfer rates of 4.5 g $dm^{-3}$ $h^{-1}$ with an energy consumption of 1$kWh^{-1}$ $kg^{-1}$ have been achieved for industrial-scale yeast production from whey using such a fermentation system.

## THE CYCLONE COLUMN

Dawson (1974) developed the cyclone column, particularly for the growth of filamentous cultures. The culture liquid was pumped from the bottom to the top of the cyclone column through a closed loop. The descending liquid ran down the walls of the column in a relatively

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thin film. Nutrients and air were fed in near the base of the column whilst the exhaust gases left at the top of the column. Good gas exchange, lack of foaming and limited wall growth has been claimed with this fermenter. Dawson (1974, 1988) has listed a number of potential bacterial, fungal and yeast applications including the batch production of a vaccine for scours in calves with the vessel being operated as batch, continuous or fed-batch.

## THE PACKED TOWER

The packed tower is a well-established application of immobilized cells. A vertical cylindrical column is packed with pieces of some relatively inert material, e.g. wood shavings, twigs, coke, an aggregate or polythene. Initially both medium and cells are fed into the top of the packed bed. Once the cells have adhered to the support and are growing well as a thin film, fresh medium is added at the top of the column and the fermented medium is removed from the bottom of the column. The best known example is the vinegar generator, in which ethanol was oxidized to acetic acid by strains of Acetobacter supported on beech shavings; the first recorded use was in 1670 (Mitchell, 1926).

More recently, packed towers have been used for sewage and effluent treatment (Noble et al., 1964). In treatment of gas liquor, a column was packed to a height 7.9 m with 'Dowpac', a polystyrene derivative. The main advantages compared with other methods of effluent treatment being its simplicity of operation and a saving in land because of the increased surface areas within the column. Other possible applications with immobilized cells are now being investigated.

## ENZYME REACTORS

### PACKED BED

Packed-bed reactors are used with immobilized or particulate biocatalysts. The reactor consists of a tube, usually vertical, packed with catalyst particles. Medium can be fed either at the top or bottom of the column and forms a continuous liquid phase between the particles. Damage due to particle attrition is minimal in packed beds compared with stirred reactors. Packed-bed reactors have been used commercially with immobilized cells and enzymes for production of aspartate and fumarate, conversion of penicillin to 6-aminopenicillanic acid, and resolution of amino acid isomers. Mass transfer between the liquid medium and solid catalyst is facilitated at high liquid flow rates through the bed; to achieve this, packed beds are often operated with liquid recycle as shown in Fig.1.

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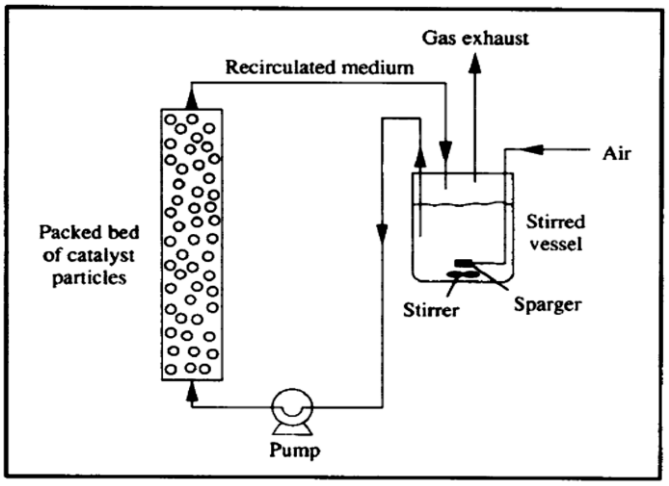



Figure 1. Packed bed reactors showing liquid recycling

The catalyst is prevented from leaving the column by screens at the liquid exit. The particles should be relatively incompressible and able to withstand their own weight in the column without deforming and occluding liquid flow. Recirculating medium must also be clean and free of debris to avoid clogging the bed. Aeration is generally accomplished in a separate vessel; if air is sparged directly into the bed, bubble coalescence produces gas pockets and flow channeling or mal-distribution. Packed beds are unsuitable for processes which produce large quantities of carbon dioxide or other gases which can become trapped in the packing.

## FLUIDIZED BED

When packed beds are operated in upflow mode with catalyst beads of appropriate size and density, the bed expands at high liquid flow rates due to upward motion of the particles. This is the basis for operation of fluidized-bed reactors as illustrated in Fig.2.

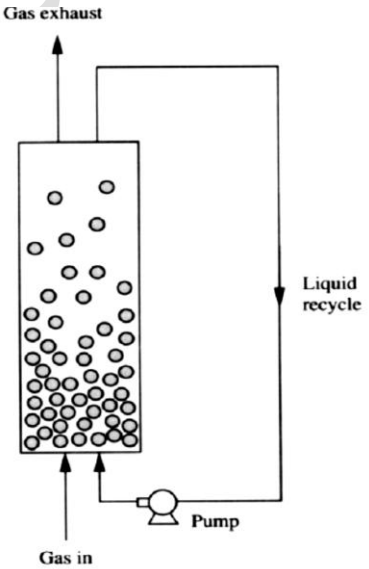



Figure 2. Packed bed reactor showing upward motion of particles.

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Because particles in fluidized beds are in constant motion, channeling and clogging of the bed are avoided and air can be introduced directly into the column. Fluidized-bed reactors are used in waste treatment with sand or similar material supporting mixed microbial populations. They are also used with flocculating organisms in brewing and for production of vinegar.

## IMMOBILIZED CELL BIOREACTORS

Cell immobilization may be achieved in one of the following ways:

- Cells may be directly bound to water insoluble carriers, e.g., cellulose, dextran, ion exchange resins, porous glass, brick, sand, etc., by adsorption, ionic bonds or covalent bonds.
- They can be cross linked to bi- or multifunctional reagents, e.g., glutaraldehyde, etc.
- Polymer matrices may be used for entrapping cells; such matrices are polyacrylamide cell, κ-carrageenan (a polysaccharide isolated from a seaweed), calcium alginate (alginate is extracted from seaweed), poly glycol oligomers, etc.

Out of these approaches, calcium alginate immobilization is the most commonly used since it can be used for even very sensitive cells. Cell immobilization has been used for commercial production of amino acids, organic acids, etc.

Bioreactors of this type are based on immobilized cells. Cell immobilization is advantageous when,

(i) the enzymes of interest are intracellular,

(ii) extracted enzymes are unstable,

(iii) the cells do not have interfering enzymes or such enzymes are easily inactivated/removed and

(iv) the products are low molecular weight compounds released into the medium.

- Under these conditions, immobilized cells offer the following advantages over enzyme immobilization:
  Enzyme purification is not needed,
- High activity of even unstable enzymes,
- High operational stability,
- Lower cost and
- Possibility of application in multistep enzyme reactions.

In addition, immobilization permits continuous operation of bioreactor, which reduces the reactor volume and, consequently, pollution problems. Obviously, immobilized cells are used for such biotransformations of compounds, which require action of a single enzyme.

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# INSTRUMENTATION IN BIOREACTORS

## BODY CONSTRUCTION

### CONSTRUCTION MATERIALS

In fermentations with strict aseptic requirements it is important to select materials that can withstand repeated steam sterilization cycles. On a small scale (1 to 30 $dm^3$) it is possible to use glass and/or stainless steel. Glass is useful because it gives smooth surfaces, is non-toxic, corrosion proof and it is usually easy to examine the interior of the vessel. Two basic types of fermenter are used:

- A glass vessel with a round or flat bottom and a top flanged carrying plate. The large glass containers originally used were borosilicate battery jars (Brown and Peterson, 1950). All vessels of this type have to be sterilized by autoclaving. Cowan and Thomas (1988) state that the largest practical diameter for glass fermenters is 60 cm.
- A glass cylinder with stainless-steel top and bottom plates. These fermenters may be sterilized in situ, but 30 cm diameter is the upper size limit to safely withstand working pressures (Solomons, 1969). Vessels with two stainless steel plates cost approximately 50% more than those with just a top plate.

At pilot and large scale, when all fermenters are sterilized in situ, any materials used will have to be assessed on their ability to withstand pressure sterilization and corrosion and on their potential toxicity and cost. Mostly used material is stainless steel: The American Iron and Steel Institute (AISI) states that steels containing less than 4% chromium are classified as steel alloys and those containing more than 4% are classified as stainless steels. The minimum amount of chromium needed to resist corrosion will depend on the corroding agent in a particular environment, such as acids, alkalis, gases, soil, salt or fresh water. Increasing the chromium content enhances resistance to corrosion, but only grades of steel containing at least 10 to 13% chromium develop an effective film. The inclusion of nickel in high percent chromium steels enhances their resistance and improves their engineering properties. The presence of molybdenum improves the resistance of stainless steels to solutions of halogen salts and pitting by chloride ions in brine or sea water. Corrosion resistance can also be improved by tungsten, silicone and other elements. AISI grade 316 steels which contain 18% chromium, 10% nickel and 2-2.5% molybdenum are now commonly used in fermenter construction.

## SEALS

At this stage it is important to consider the ways in which a reliable aseptic seal is made between glass and glass, glass and metal or metal and metal joints such as between a fermenter vessel and a detachable top or base plate. With glass and metal, a seal can be made with a compressible

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gasket, a lip seal or an 'O' ring .With metal to metal joints only an 'O' ring is suitable. This is placed in a groove, machined in either the end plate, the fermenter body or both. This seal ensures that a good liquid- and/or gas-tight joint is maintained in spite of the glass or metal expanding or contracting at different rates with changes in temperature during a sterilization cycle or an incubation cycle. Nitryl or butyl rubbers are normally used for these seals as they will withstand fermentation process conditions.

The structural components of the fermenter involved in aeration and agitation are:

- The agitator (impeller)
- Stirrer glands and bearings
- Baffles
- The aeration system (sparger)

## THE AGITATOR (IMPELLER)

The agitator is required to achieve a number of mixing objectives, e.g. bulk fluid and gas-phase mixing, air dispersion, oxygen transfer, heat transfer, suspension of solid particles and maintaining a uniform environment throughout the vessel contents. Agitators may be classified as disc turbines, vaned discs, open turbines of variable pitch and propellers.

## STIRRER GLANDS AND BEARINGS

The satisfactory sealing of the stirrer shaft assembly top plate has been one of the most difficult problems to overcome in the construction of fermentation equipment which can be operated aseptically for long periods.

Four basic types of seal assembly have been used: the stuffing box (packed-gland seal), the simple bush seal, the mechanical seal and the magnetic drive. Most modern fermenter stirrer mechanisms now incorporate mechanical seals instead of stuffing boxes and packed glands. Mechanical seals are more expensive, but are more durable and less likely to be an entry point for contaminants or a leakage point for organisms or products which should be contained. Magnetic drives, which are also quite expensive, are being used in some animal cell culture vessels.

## BAFFLES

Four baffles are normally incorporated into agitated vessels of all sizes to prevent a vortex and to improve aeration efficiency. In vessels, over $3dm^3$ diameter six or eight baffles may be used. Baffles are metal strips roughly one-tenth of the vessel diameter and attached radially to the wall.

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## THE AERATION SYSTEM (SPARGER)

A sparger may be defined as a device for introducing air into the liquid in a fermenter. Three basic types of sparger have been used and may be described as the porous sparger, the orifice sparger (a perforated pipe) and the nozzle sparger (an open or partially closed pipe).

## VALVES AND STEAM TRAPS

Valves attached to fermenters and ancillary equipment is used for controlling the flow of liquids and gases in a variety of ways.

The valves may be:

- Simple ON/OFF valves which are either fully open or fully closed.
- Valves, which provide coarse control of flow rates.
- Valves, which may be adjusted very precisely so that flow rates may be accurately controlled.
- Safety valves that are constructed in such a way that liquids or gases will flow in only one direction.

### GATE VALVES

In this valve, a sliding disc is moved in or out of the flow path by turning the stem of the valve. It is suitable for general purposes on a steam or a water line for use when fully open or fully closed and therefore should not be used for regulating flow. The flow path is such that the pressure drop is minimal, but unfortunately it is not suitable for aseptic conditions as mash solids can pack in the groove where the gate slides, and there may be leakage round the stem of the valve which is sealed by a simple stuffing box. This means that the nut around the stem and the packing must be checked regularly.

### GLOBE VALVES

In this valve, a horizontal disc or plug is raised or lowered in its seating to control the rate of flow. This type of valve is very commonly used for regulating the flow of water or steam since it may be adjusted rapidly. It is not suitable for aseptic operation because of potential leakage round the valve stem, which is similar in design to that of the gate valve. There is a high-pressure drop across the valve because of the flow path.

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## PISTON VALVES

Piston valve is similar to a globe valve except that the flow is controlled by a piston passing between two passing rings.

## PLUG VALVES

In this valve there is a parallel or tapered plug sitting in a housing through which an orifice, A, has been machined. When the plug is turned through 90° the valve is fully open and the flow path is determined by the cross-sectional area of A, which may not be as large as that of the pipeline. This type of valve has a tendency to leak or seize up, but the use of lubricants and/or sealants may overcome these problems. If suitable packing sleeves (e.g. compressed asbestos) are incorporated into the valve it will be suitable for use in a steam line as it is quick to operate, has protected seals, a minimal pressure drop and a positive closure. It can also provide good flow control.

## BALL VALVES

This valve has been developed from the plug valve. The valve element is a stainless-steel ball through which an orifice is machined. The ball is sealed between two wiping surfaces which wipe the surface and prevent deposition of matter at this point. The orifice in the ball can be of the same diameter as the pipeline, giving an excellent flow path. The valve is suitable for aseptic operation, can handle mycelial broths and can be operated under high temperatures and pressures. The pressure and temperature range is normally limited by the PTFE seat and stem seals.

## BUTTERFLY VALVES

The butterfly valve consists of a disc which rotates about a shaft in housing. The disc closes against a seal to stop the flow of liquid. This type of valve is normally used in large diameter pipes operating under low pressure where absolute closure is not essential. It is not suitable for aseptic operation.

## PINCH VALVES

In the pinch valve, a flexible sleeve is closed by a pair of pinch bars or some other mechanism, which can be operated by compressed air remotely or automatically. The flow rate can be controlled from 10 to 95% of rated flow capacity (Pikulik, 1976). The valve is suitable for aseptic operation with fermentation broths, even when mycelial, as there are no dead spaces in

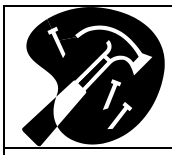
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the valve structure, and the closing mechanism is isolated from the contents of the piping. Obviously, the sleeve of rubber, neoprene, etc., must be checked regularly for signs of wear.

## DIAPHRAGM VALVES

Like the pinch valve, the diaphragm valve makes use of a flexible closure, with or without a weir. It may also be fitted with a quick action lever. This valve is very suitable for aseptic operation provided that the diaphragm is of a material which will withstand repeated sterilization. The valve can be used for ON/OFF, flow regulation, and for steam services within pressure limits. Diaphragm failure, which is often due to excessive handling, is the primary fault of the valve.

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## GATE Questions

GATE 2010

Q.1 In a chemostat operating under steady state, a bacterial culture can be grown at dilution rate higher than maximum growth rate by

(A) partial cell recycling (B) using sub-optimal temperature

(C) pH cycling (D) substrate feed rate cycling

Q.2 An immobilized enzyme being used in a continuous plug low reactor exhibits an effectiveness factor ($\eta$) of 1.2. The value of $\eta$ being grater than 1.0 could be apparently due to :

(A) substrate inhibited kinetics with internal pore diffusion limitation

(B) external pore diffusion limitation

(C) sigmoidal kinetics

(D) unstability of the enzyme

Q.3 A roller bottle culture vessel perfectly cylindrical in shape having inner radius (r) = 10cm and length (l) = 20cm was fitted with a spiral film of length (L) = 30 cm and width (W) = 20 cm. If the film can support $10^3$ anchorage dependent cells per $cm^2$ , the increase in the surface area after fitting the spiral film and the additional number of cells that can be grown respectively are :

(A) 1200$cm^2$ and $12\times10^7$ cells (B) 600 $cm^2$ and $6\times10^7$ cells

(C) 600 $cm^2$ and 8300 cells (D) 1200 $cm^2$ and 8300 cells

GATE 2011

Q. 4 In balanced growth phase of a cell

P. a high degree of lateral branching

Q. genetic instability of culture

R. an absence of geotropism

S. poor biomass production

(A) P and R only (B) P and Q only

(C) Q and R only (D) R and S only

Q.5 Substrate consumption in lag phase of microbial growth is primarily used for

P. turnover of cell material

Q. maintenance of intracellular pH

R. motility

S. increase in cell number

(A) P,Q and S only (B) Q,R and S only

(C) P,Q and R only (D) S only

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Q.6 Wash out (as defined by D= $\mu_{max}$) of continuous stirred tank fermenter is characterized by (X=biomass, S= substrate concentration in bioreactor, $S_0$=substrate concentration in feed, P=product concentration in bioreactor)

(A) X = 0, S = 0, P = 0 (B) X = 0, S = $S_0$ , P = 0

(C) X = 0, S < 0, P = 0 (D) X < 0, S < 0, P < 0

Q.7 Match the microbial growth characteristics in Group I with the corresponding features in Group II

| Group I | Group II |
|---|---|
| P. Growth associated product formation | 1. Specific growth rate decreases with increasing product concentration |
| Q. Non growth associated product | 2.Specific product formation rate is constant formation |
| R. Product Inhibition | 3.Specific product formation rate is proportional to specific growth rate |
| S.Substrate inhibition | 4. Specific growth rate decreases with increasing substrate concentration |

(A) P-1,Q-2, R-4 , S-3 (B) P-3,Q-2, R-1 , S-4

(C) P-2,Q-1, R-3 , S-4 (D) P-2,Q-3, R-4 , S-1

Q.8 Maximum specific growth rate ($\mu_{max}$) of a microorganism is calculated by taking the (ln = $\log_e$, X = biomass, t = time)

(A) slope of ln X vs t of the growth cycle

(B) slope of ln X vs t during the exponential growth phase

(C) slope of X vs t

(D) slope of X vs t during the exponential phase of growth

**Common Data for Questions 9 and 10:**

A microorganism grows in a continous chemostat culture of 60$m^3$ working volume with sucrose as the growth limiting nutrient at dilution rate, D = 0.55$h^{-1}$. The steady state biomass concentration is 4.5 Kg dry biomass $m^{-3}$ and the residual sucrose concentration is 2.0 $Kgm^{-3}$.The sucrose concentration in the incoming feed medium is 10.0 $Kgm^{-3}$.

Q.9 What would be the yield $Y_{X/S}$(Kg biomass/Kg substrate)?

(A) 0.562 (B) 0.462 (C) 0.362 (D) 0.162

Q.10 What would be the sucrose concentration in the input feed for the output to be 45Kg biomass $h^{-1}$ ?

(A) 3.225Kg $m^{-3}$ (B) 4.425 Kg $m^{-3}$ (C) 5.115Kg $m^{-3}$ (D) 6.525Kg $m^{-3}$

Q.11 In an exponentially growing batch culture of *Saccharomyces cerevisiae*, the cell density is 20 $gl^{-1}$(DCW), the specific growth rate (μ) is 0.4 $h^{-1}$and substrate uptake rate(v) is 16 $gl^{-1}h^{-1}$. The cell yield coefficient $Y_{x/s}$ will be (2012)

(A) 0.32 (B) 0.64 (C) 0.80 (D) 0.50

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GATE 2012

Q.12 Determine the correctness or otherwise of the following Assertion (a) and Reason (r).
Assertion: Cell mass yield of methylotrophic yeast is more on methanol compared to glucose.
Reason: Methanol has a greater degree of reductance compared to glucose.
(A) Both (a) and (r) are correct and (r) is the correct reason for (a)
(B) (a) is correct, (r) is false
(C) (a) is false, (r) is correct
(D) Both (a) and (r) are correct but (r) is not the correct reason for (a)

Answers:

| | | | | | |
|---|---|---|---|---|---|
| Q.1 – D | Q.2 – A | Q.3 – B | Q.4 – C | Q.5 – C | Q.6 – B |
| Q.7 – B | Q.8 – B | Q.9 – A | Q.10 – B | Q.11 – D | Q.12 - A |

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